

Targeting the intrinsically disordered AR-NTD through a machine learning-based enhanced sampling workflow

Received: 10 October 2025

Accepted: 18 May 2026

Published online: 05 June 2026

 Check for updates

Kai Zhu^{1,2,3,4}, Huating Wang^{1,2,3,4}, Jintu Zhang^{1,2,3,4}, Renling Hu^{1,2,3},
Linlong Jiang^{1,2,3}, Hui Zhang^{1,2,3}, Yu Kang^{1,2,3}, Tingjun Hou^{1,2,3}✉ &
Dan Li^{1,2,3}✉

Targeting the intrinsically disordered N-terminal domain of the androgen receptor (AR-NTD) represents a promising strategy to overcome resistance in prostate cancer. However, its inherent lack of a stable tertiary structure and highly dynamic conformational ensemble pose formidable challenges for rational drug design. This study introduces an integrated computational workflow that combines enhanced sampling techniques and machine learning collective variables to identify druggable conformations of the AR-NTD and elucidate the binding mechanism of its modulator, EPI-002. We characterize nine metastable states of the Tau-5 region and reveal that ligand recognition is driven by π - π stacking and structured water-mediated hydrogen bonds. Leveraging these insights, we perform structure-based virtual screening based on the identified druggable conformations and identify K53, a rationally designed AR-NTD antagonist, which exhibits potent anti-proliferative activity in enzalutamide-resistant prostate cancer cells. K53 directly binds the AR-NTD, suppresses AR transcriptional activity, and demonstrates high selectivity for cancer cells. This work provides a rational design paradigm for targeting intrinsically disordered proteins and offers a therapeutic candidate for resistant prostate cancer.

Intrinsically disordered proteins (IDPs) play diverse biological roles and are closely related to a wide range of major human diseases^{1–3}. Unlike folded proteins, IDPs lack a stable three-dimensional (3D) structure under physiological conditions due to their relatively flat free energy landscapes, resulting in highly heterogeneous conformational ensembles. Consequently, describing IDPs using a single or limited number of representative structures is insufficient to capture their structural complexity. Instead, dynamic conformational ensembles, representing a distribution of interconverting states, could better capture their structural heterogeneity and flexibility. This strategy has been applied across multiple systems, including α -synuclein⁴, tau⁵,

c-Myc⁶, and p53⁷, demonstrating its utility in IDP-targeted drug discovery. However, these studies have not yet provided a comprehensive exploration of IDP conformation space, nor have they systematically investigated IDP-ligand interactions to guide rational drug design, highlighting the need for advanced computational and experimental workflows to systematically map conformational landscapes and decode ligand-binding mechanisms, thereby accelerating the development of IDP-targeted therapeutics.

Enhanced sampling techniques in molecular dynamics (MD) simulations represent powerful tools to characterize IDP ensembles. These approaches enable extensive exploration of conformational

¹College of Pharmaceutical Sciences, Zhejiang University, Hangzhou, Zhejiang, China. ²Jinhua Institute of Zhejiang University, Jinhua, Zhejiang, China.

³Zhejiang Provincial Engineering Research Center for Innovative Small-Molecule Drugs, Jinhua, China. ⁴These authors contributed equally: Kai Zhu, Huating Wang, Jintu Zhang. ✉e-mail: tingjunhou@zju.edu.cn; lidancps@zju.edu.cn

space and reconstruction of free energy landscapes, yielding both representative conformations and their statistical weights. Enhanced sampling techniques can be broadly classified into two types: collective variable (CV)-based methods and CV-free methods⁸, the distinction primarily reflects whether explicit low-dimensional CVs are introduced. CV-based techniques apply bias potential or biasing force along predefined CVs to facilitate barrier crossing⁹, with prominent examples including umbrella sampling¹⁰, metadynamics^{11,12}, adaptive biasing force¹³, and on-the-fly probability enhanced sampling (OPES)^{14–16}. In contrast, CV-free methods without relying on predefined CVs, typically by modifying the effective sampling distribution or potential energy surface. Representative examples include replica exchange molecular dynamics (REMD)^{17,18}, replica exchange with solute tempering (REST)^{19,20}, and accelerated molecular dynamics (aMD)²¹ enhance sampling by constructing a series of overlapping probability distributions without relying on predefined CVs.

The conformational dynamics of IDPs are often high-dimensional, diverse, and mutually orthogonal, posing significant challenges in defining low-dimensional CVs that accurately capture their slow dynamical modes. As a result, CV-free approaches like REST have been adopted in IDP studies due to their simplicity and generality^{22–24}. While REST provides a global, unbiased enhancement of sampling, it still lacks the ability to selectively probe how ligands and proteins actually associate. In contrast, the emergence of machine learning-based collective variables (MLCVs)²⁵ has introduced new opportunities, as they not only retain sampling efficiency but also enable a more directed characterization of ligand–protein binding pathways. MLCVs can be categorized into three primary categories: supervised, unsupervised, and time-informed learning. Supervised methods, such as Deep Linear Discriminant Analysis (Deep-LDA)²⁶ and Deep Targeted Discriminant Analysis (Deep-TDA)²⁷, aim to distinguish known metastable states. Unsupervised learning approaches, such as AutoEncoder-based CVs (AE CVs)^{28,29}, focus on dimensionality reduction. Time-informed learning methods, exemplified by Time-lagged Independent Component Analysis (Deep-TICA)³⁰, are designed to extract the system's slow dynamical modes. A series of studies has demonstrated the considerable potential of MLCVs in biomolecular simulations^{31–34}, providing a data-driven framework to capture complex conformational dynamics and enhance sampling efficiency in disordered systems.

The N-terminal domain of the androgen receptor (AR-NTD), a prototypical IDP that functions as the AR's primary transactivation domain, plays an essential role in prostate cancer (PCa)³⁵. Currently, all androgen receptor (AR) antagonists approved for PCa treatment, such as bicalutamide and enzalutamide (Enz), share a primary mechanism of action: targeting the ligand-binding pocket of the AR ligand-binding domain (AR-LBD). However, the majority of patients develop acquired resistance to these therapies. Clinical resistance mechanisms commonly include AR-LBD mutations, AR gene amplification, the expression of AR splice variants (ARVs), and de novo intratumoral synthesis of androgens³⁶. It is particularly noteworthy that ARVs, which lack the ligand-binding pocket or entire LBD, are inherently resistant to both marketed AR-LBD-targeted antagonists and those in development. ARVs are expressed in over 20% of early-stage PCa patients and nearly 40% of castration-resistant prostate cancer (CRPC) patients³⁷, posing significant therapeutic challenges. Given that the resistance mechanisms of mutations, amplification, and splice variants all exert their effects through the AR signaling pathway while preserving the integrity of the NTD, the development of AR antagonists targeting the AR-NTD represents a promising strategy to overcome clinical resistance, particularly against ARVs that evade current LBD-directed therapies.

The primary compounds targeting the AR-NTD are EPI-001 and its analogs, which were discovered through high-throughput screening and are derivatives of bisphenol A diglycidyl ether (BADGE) originally isolated from marine sponges, and they represent promising candidate therapeutics for CRPC³⁸. However, EPI analogs suffer from several

limitations: modest binding affinity, limited specificity toward the AR-NTD, and a covalent binding mechanism that may lead to off-target reactivity and toxicity concerns. Importantly, medicinal chemistry optimization of the EPI scaffold yielded EPI-7386³⁹, a nonvalent analog lacking the epoxide warhead with approximately 20-fold higher potency than EPI-001 that progressed to clinical evaluation. However, masfaniten ultimately failed in phase I/II trials, highlighting the ongoing challenges of achieving robust efficacy and selectivity when targeting the intrinsically disordered AR-NTD. Therefore, elucidating how EPI analogs recognize the binding site of the AR-NTD is critical, not only for understanding their mode of action but also for enabling the rational, structure-guided design of next-generation AR-NTD antagonists with improved efficacy and safety. It has been reported that EPI analogs covalently bind to Cys404 at the Tau-5 region of the AR-NTD^{40,41}. Previous enhanced sampling studies have primarily examined the interaction of EPI-002, the most potent stereoisomer of EPI-001, with the R2–R3 subregions of Tau-5. These studies observed that the ligand engages transient helical surfaces, induces partially folded rotational states, and that aromatic residues predominantly drive this process¹⁶. However, the covalent nature of EPI-002 remains poorly characterized. Recently, MD studies of covalent EPI-002 adducts have shown that covalent attachment can markedly alter the conformational ensemble of the AR-NTD⁴². Nevertheless, it remains unclear how small molecules initially engage the AR-NTD and progress toward covalent adduct formation, knowledge that would be essential for guiding rational drug design.

In this work, we established a generalized workflow integrating CV-free enhanced sampling, CV-based enhanced sampling, and MLCVs to identify druggable conformations of IDPs and elucidate their ligand interaction mechanisms, thereby providing guidance for rational drug design. As a case study, this workflow is applied to investigate the AR-NTD. The analysis is expanded to the full Tau-5 ensemble, explicitly characterizing how transient binding pockets guide EPI-002 recognition and facilitate its covalent stabilization at Cys404. Our findings reveal that the hydrogen-bond networks between EPI-002 and the protein/water environment essentially reinforce its π – π stacking with aromatic residues, thereby underpinning ligand recognition and covalent locking. Furthermore, virtual screening based on its workflow identified an antagonist, K53, the first rationally designed antagonist against the AR-NTD. Its discovery highlights not only the conformational plasticity of the AR-NTD that underpins its druggability but also the practical feasibility of our strategy for elucidating ligand recognition in IDPs and discovering lead compounds targeting IDPs.

Results

An ML-based enhanced sampling workflow for deciphering ligand binding mechanisms to IDPs

Our workflow is illustrated in Fig. 1. Given the high-dimensional complexity and orthogonal dynamics of IDPs, we first employ the REST algorithm to explore the conformational space. REST3⁴³ modulates the strength of van der Waals solvent–solvent interactions, thereby alleviating the severe conformational collapse commonly observed in larger IDPs²⁰, which would otherwise hinder extensive conformational sampling.

After sampling the conformational space of the system by REST, the next critical step is to identify druggable states. From a probabilistic perspective, ligand binding is expected to occur preferentially within metastable conformational states. To extract these states, we adopt AEs as the dimensionality reduction strategy. Specifically, we perform AE-based dimensionality reduction on the ambient-temperature replica trajectory from REST3 simulations. Due to the temporal discontinuity inherent in replica exchange trajectories (resulting from frequent swapping between replicas), conventional time-informed methods cannot be directly applied. AEs, however, offer a robust alternative. AE is a neural network composed of two key

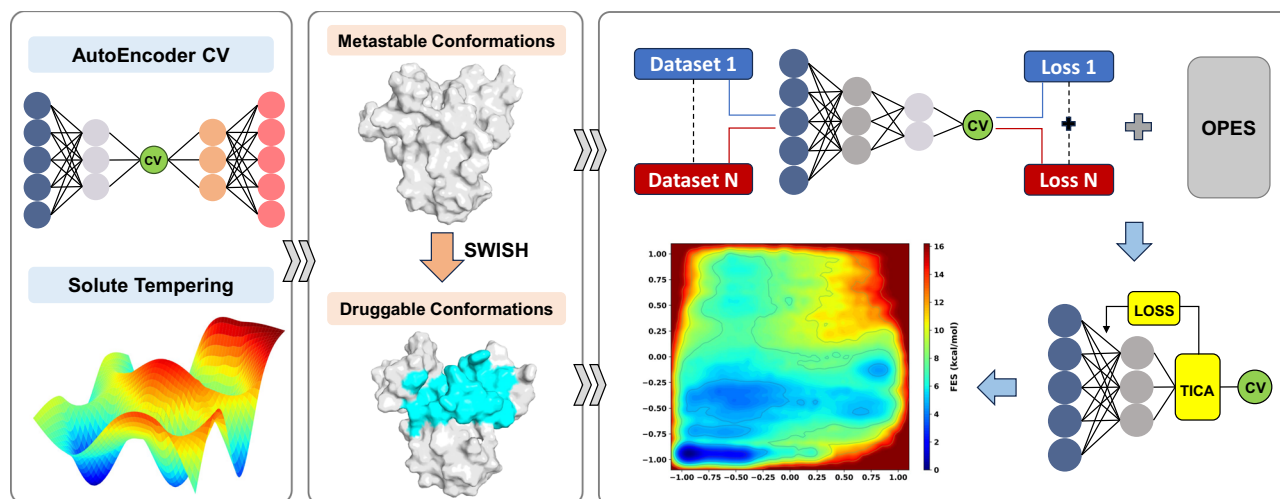


Fig. 1 | Machine learning-based enhanced sampling workflow for identifying druggable conformations of IDPs. REST3 simulations are first used to enhance conformational sampling of IDPs, and AE CVs are then applied to identify metastable conformational states. SWISH simulations with small-molecule solvent probes are used to reveal cryptic ligand-binding pockets within these metastable

conformations. The identified pockets are subsequently used for molecular docking and ligand–protein binding analysis. Finally, multitask learning CVs are combined with OPES to calculate binding free energies, and Deep-TICA is used to extract slow CVs and characterize ligand-binding and unbinding mechanisms.

components: an encoder E , which maps high-dimensional inputs to a lower-dimensional latent space, and a decoder D , which reconstructs the original input from this latent representation⁴⁴. The loss function of AE is defined as

$$\mathcal{L}_{\text{AE}} = \sum_{i=1}^N |x_i - D \circ E(x_i)|^2 \quad (1)$$

where x_i represents the i -th input sample. The bottleneck layer of the trained AE serves as the CV in subsequent analyses.

While metastable conformations can be identified from the AE-transformed landscape, transient binding pockets often remain concealed. To reveal these cryptic sites, we employ the Sampling Water Interfaces through Scaled Hamiltonians (SWISH)⁴⁵ method, which combines small-molecule solvent probes with parallel tempering. By perturbing protein–water interactions, SWISH effectively mimics the presence of small-molecule ligands and reveals otherwise cryptic hydrophobic pockets, thereby exposing hidden hydrophobic pockets and making it particularly suitable for probing druggability in disordered systems.

Once potential binding pockets are identified, known small molecules can be docked into these sites. To further characterize the binding mechanism and estimate free energy, enhanced sampling is required. Here, the presence of well-defined bound and unbound states enables the construction of CVs via multitask learning, which integrates information from both labeled and unlabeled datasets. Specifically, we integrate Deep-TDA²⁷ and AutoEncoders with a unified multitask learning framework⁴⁶.

In Deep-TDA, metastable states are modeled as a mixture of N_s multivariate Gaussian distributions with diagonal covariances, characterized by $N_\rho = N_s - 1$ CV centers and associated covariances. The loss function is defined as:

$$\mathcal{L}_{\text{Deep-TDA}} = \sum_k^{N_s} \sum_\rho^{N_\rho} \left(L_{k,\rho}^\mu + \alpha L_{k,\rho}^\sigma \right) \quad (2)$$

where μ and σ denote the mean and variance terms. The total multitask loss combines the AE reconstruction loss (from unlabeled data) and

the Deep-TDA loss (from labeled data) as

$$\mathcal{L}_{\text{multitask}} = \mathcal{L}_{\text{AE}} + \alpha \mathcal{L}_{\text{Deep-TDA}} \quad (3)$$

Short conventional MD (cMD) simulations of the bound and unbound states provide the labeled dataset, while enhanced sampling trajectories of ligand dissociation serve as unlabeled data. This multitask approach enables simultaneous learning of metastable and transition-state features, yielding efficient and physically meaningful CVs. Compared with time-informed approaches, which require long and continuous trajectories, this strategy is more practical for disordered systems.

For the enhanced sampling itself, we employ OPES¹⁴, an advanced method that constructs a bias potential $V(s)$ by dynamically estimating the system's probability distribution and guiding sampling toward a target distribution $P_{\text{tg}}(s)$.

$$V(s) = -\frac{1}{\beta} \log \frac{P_{\text{tg}}(s)}{P(s)} \quad (4)$$

where $\beta = 1/(k_B T)$, $P(s)$ is the equilibrium probability distribution. Compared to well-tempered metadynamics, OPES offers improved control over the target distribution, reduced sensitivity to hyperparameter choices, and a more direct and efficient recovery of FES. Among its variants, OPES-explore¹⁶ is effective in for exploring complex free energy landscapes, despite its relatively slower convergence in recovering accurate thermodynamics.

To further refine CVs and enable a thermodynamic and structural characterization of the system, we extract the system's slow modes using Deep-TICA³⁰, a deep learning extension of Time-lagged Independent Component Analysis (TICA)⁴⁷ grounded in the Variational Approach for Conformational dynamics (VAC)^{48–51}. Deep-TICA learns nonlinear transformations that maximize the time autocorrelation of the projected dynamics. The Deep-TICA loss function is given by:

$$\mathcal{L}_{\text{Deep-TICA}} = -\sum_{i=1}^n \lambda_i^2 \quad (5)$$

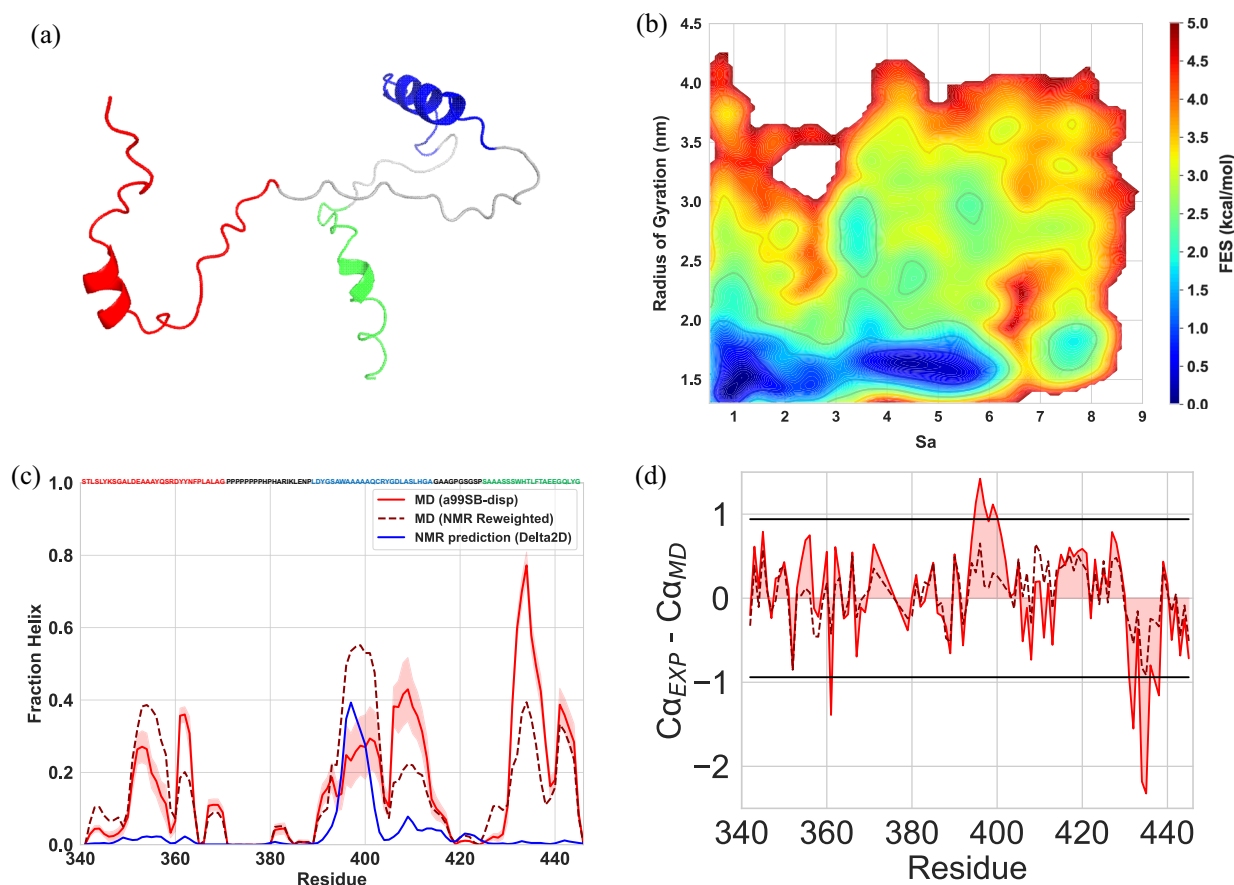


Fig. 2 | Enhanced sampling of the Tau-5 region via replica exchange with REST3.

a AF2-predicted initial structure of Tau-5. R1 is highlighted in red, R2 in blue, R3 in green, and the remaining residues in gray. **b** Free energy surface (FES) of the 300 K replica trajectory projected onto the radius of gyration (Rg) and the α -helical order parameter (Sa). **c** Helical propensity of Tau-5 obtained from the REST3 MD simulations compared with the experimental NMR data. The DSSP-calculated helicity from the MD trajectories (red) and maximum entropy reweighted ensemble constrained by NMR C α chemical shifts (dashed dark red) are compared against

Delta2D-derived helicity from experimental chemical shifts (blue). Data from MD trajectories are presented as mean values with error bands representing the standard deviation across trajectory segments. **d** Experimental NMR C α chemical shifts versus simulated values before (red) and after (dashed dark red) reweighting. Black lines represent the average C α chemical shift prediction error of SPARTA+ evaluated on its training dataset of folded protein structures. Data from MD trajectories are presented as mean values with error bands representing the standard deviation across trajectory segments.

where λ_i are the top n eigenvalues of the learned transfer operator. Since Deep-TICA relies on VAC theory, it requires continuous, equilibrated, and reversible trajectories for proper learning.

Harvesting metastable conformations of the Tau-5 of AR-NTD Using REST3 and AE CV

In our study, we selected the Tau-5 fragment of the AR-NTD and its ligand EPI-002 as a model system. Circular dichroism (CD) experiments and NMR secondary chemical shifts have shown that Tau-5 exhibits ~10, 30, and 10% helical content in regions R1, R2, and R3, respectively^{40,52–54}. To initiate MD simulations, we predicted the 3D structure of the 106-residue Tau-5 fragment (S341-G446) using AlphaFold2 (AF2)⁵⁵. As shown in Fig. 2a, the predicted structure exhibits helical contents of 19.4, 41.7, and 28.6% for R1, R2, and R3, respectively, in reasonable agreement with experimental measurements, thereby validating the AF2 structure as a suitable starting point for subsequent simulations.

To sample the conformational ensemble of Tau-5, we performed MD simulations using the a99SB-disp protein force field and the corresponding a99SB-disp water model⁵⁶. Enhanced sampling was carried out using replica exchange with REST3, employing eight replicas distributed over 300–450 K, with all protein atoms selected for solute tempering. Replica exchange attempts were performed every 2 ps,

resulting in an average exchange acceptance rate of approximately 6.5%, which is sufficient to ensure replica mixing and enable exploration of the conformational space. Full simulation parameters are provided in Supplementary Table 1. Each replica was simulated for 4.1 μ s, yielding a total aggregated simulation time of 32.8 μ s.

Convergence of the 300 K replica was evaluated using blocking analysis^{57,58} (see Methods). In addition, the time evolution of the radius of gyration (Rg) across replicas was monitored and showed stable fluctuations without systematic drift (Supplementary Fig. 1), indicating a well-equilibrated and converged ensemble. Secondary structure propensities were analyzed on the FES spanned Rg and the α -helical order parameter (Sa)⁵⁹ (Fig. 2b). The average helical propensity calculated using the DSSP algorithm is shown in Fig. 2c. Compared with the experimental data derived from the Delta2D⁶⁰ analysis of chemical shifts^{40,61}, the simulated helical content for R1 and R3 was overestimated, whereas R2 showed underestimation in its N-terminal half and overestimation in its C-terminal half.

It should be noted that secondary-structure estimates from chemical shifts (e.g., Delta2D) may be subject to sequence- and system-specific errors. For structural elements with < 30% helicity (e.g., R1 and R3), CD spectroscopy often exhibits larger deviations⁶⁰. Therefore, the apparent overestimation of R1 and R3 helicity in Delta2D analyses may not fully capture the true ensemble behavior. To further validate

simulation accuracy, we used SPARTA+⁶² to compute the NMR chemical shifts from the MD trajectory and compared them with the experimental values^{40,61} (see Fig. 2d and Table 1). The root mean square deviation (RMSD) of 0.652 ppm between computed and experimental C α chemical shifts ranks among the lowest reported in recent large-scale benchmarks across seven modern force fields for nine IDPs^{56,63}. Notably, our system (106 residues) is significantly larger than those in prior benchmarks, highlighting the robustness of our approach. Except for the residues 361, 395–397, 399, 423, 434–435, and 438, which showed slightly elevated deviations, all the other residues fell within SPARTA+'s expected prediction error of 0.92 ppm.

To further refine the ensemble, we applied a maximum entropy reweighting algorithm^{64,65} using C α chemical shifts from the 300 K trajectory as constraints ("Methods"). This correction improved agreement with experimental trends (Fig. 2c, d) and ensured that all predicted chemical shifts fell within error bounds, confirming the reliability of REST3 sampling for capturing Tau-5 conformational ensembles.

The FES projected onto Rg and S α (Fig. 2b) revealed that the dominant free energy minima reside at Rg < 2.0 nm, suggesting that metastable conformations of Tau-5 favor compact states. This observation is consistent with our hypothesis that compact IDP

conformations are more likely to form transient binding pockets. In particular, the region defined by $3 < S\alpha < 6$ and $1.4 < Rg < 2.0$ nm corresponds to the "molten globule" region previously reported²². While Rg and S α are physically interpretable, they provide limited resolution for distinguishing metastable states. To overcome this limitation, we introduced AE CVs.

We constructed the AE CVs^{28,29} using 1,899 C α –C α pairwise distances from the R1–R2, R2–R3, and R1–R3 regions as the input descriptors. The AE was trained on the 300 K REST3 trajectory, and the resulting FES projected onto the two AE CVs revealed nine distinct metastable states with pronounced structural differences (Fig. 3). The conformations within each FES minimum were structurally homogeneous, confirming the robustness and discriminative power of the learned CVs (Supplementary Fig. 2). To further investigate which descriptors contributed most to the AE CV, we conducted a sensitivity analysis of these features²⁶. The top 20 most influential descriptors are shown in Supplementary Fig. 3, with aromatic residues prominently represented. Among them, two descriptors involved aromatic–aromatic interactions, while nine corresponded to aromatic–non-aromatic contacts. Notably, HIS413 and TRP433 appeared frequently across multiple descriptors, suggesting that these residues play key roles in the conformational dynamics and metastability of Tau-5.

Table 1 | The consistency between the NMR chemical shifts calculated from the 300 K replica of the REST3 MD simulation and those obtained from the maximum entropy reweighted ensemble derived using C α chemical shifts as constraints, compared to the experimental NMR chemical shifts

	C α	HN	N	C'	CB
Tau-5 REST3 300 K Replica Ensemble					
RMSD (ppm)	0.652	0.229	1.078	0.584	0.505
Correlation	0.981	−0.583	0.845	0.811	0.811
C α -reweighted Maximum Entropy Tau-5 Ensemble					
RMSD (ppm)	0.344	0.189	0.943	0.424	0.377
Correlation	0.995	−0.083	0.964	0.907	0.999

Unveiling the binding pockets in metastable conformations using SWISH

In our SWISH simulations, benzene was used as a solvent probe at a concentration of 1 M. For each metastable conformation identified from the AE CV, four independent replicas were prepared with scaling factors λ set to 1, 1.05, 1, and 1.15. Increasing λ enhances the solvation of apolar surface regions, thereby modulating hydrophobic interactions on the protein surface and effectively mimicking the presence of small-molecule ligands.

Each metastable conformation was subjected to 50 ns of SWISH simulation. Analyses focused on the $\lambda=1.0$ replicas to identify conformations harboring druggable binding pockets. As illustrated in Fig. 4 and Supplementary Fig. 4, SWISH successfully revealed binding

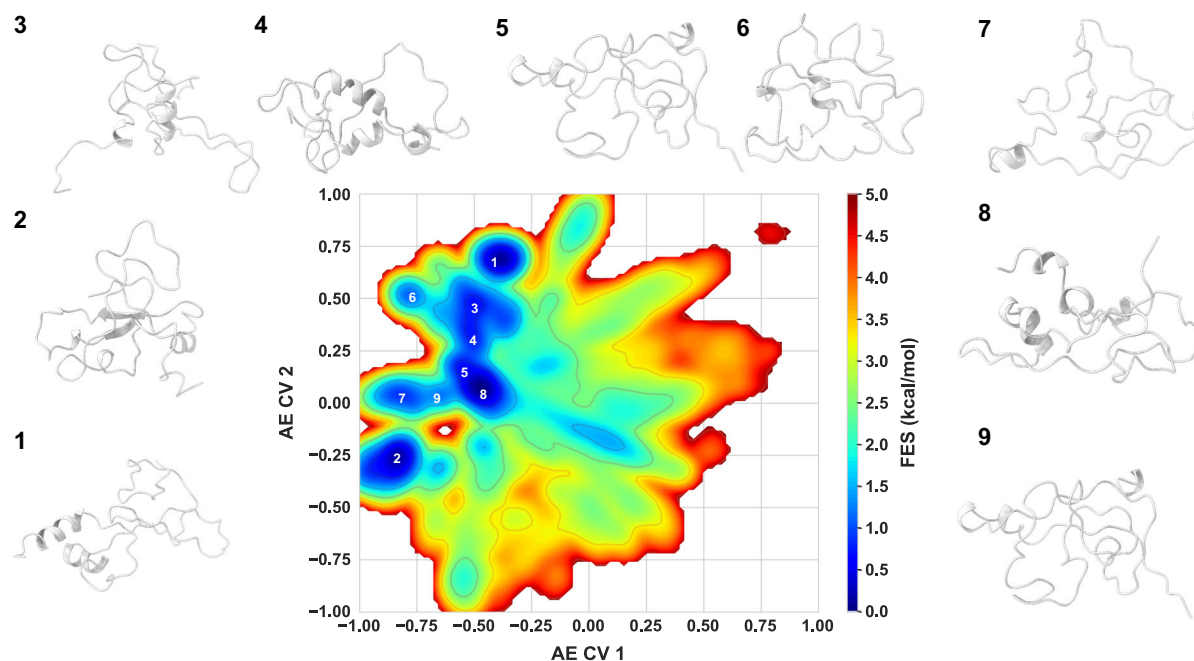


Fig. 3 | Identification of Tau-5 metastable conformations using autoencoder collective variables. The FES projected onto two AE CVs reveals nine distinct metastable states of Tau-5. Representative three-dimensional structures from each

minimum are shown to illustrate the conformational diversity captured by the learned CVs.

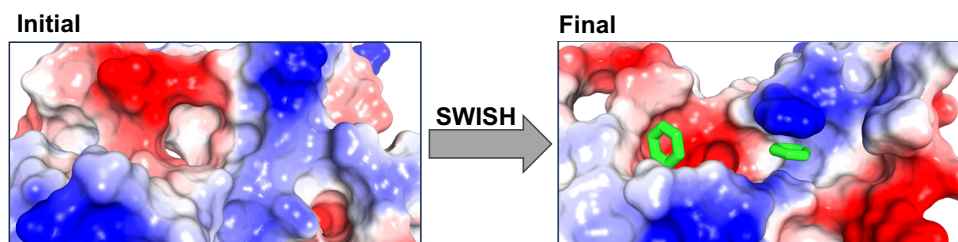


Fig. 4 | SWISH simulations uncover druggable pockets in metastable Tau-5 conformations. Representative initial and final structures from SWISH simulations show the opening of cryptic ligand-binding pockets in the metastable conformations identified from the AE CVs.

sites across all sampled metastable states, demonstrating its effectiveness in uncovering cryptic ligand-binding pockets in disordered systems.

Subsequently, molecular docking of EPI-002 into the identified pockets was performed using the Glide algorithm^{66,67}. The docking results consistently placed the chlorine atoms of EPI-002 in close proximity to the sulfur atom of Cys404 in Tau-5, strongly suggesting a favorable geometry for covalent bond formation. In addition, numerous aromatic residues were found surrounding the ligand within the ligand-binding site, implicating them in π - π stacking and other stabilizing non-covalent interactions during ligand recognition (Supplementary Fig. 5).

To further confirm the feasibility of covalent binding, we performed covalent docking^{68,69}, which confirmed that Cys404 within the identified binding pockets can indeed form covalent bonds with EPI-002. Remarkably, the covalently docked poses closely resemble those obtained from the non-covalent docking (Supplementary Fig. 6), lending further support to the hypothesized binding mode of EPI-002 with the AR-NTD.

Coupling OPES with MLCVs for precise binding free energy calculations

Water plays a pivotal role in ligand binding^{70,71} and the conformational dynamics of IDPs⁷². Inspired by recent advances in the integration of water-mediated interactions into CVs^{32,33}, we designed an MLCV to capture water behavior relevant to both ligand solvation and binding site hydration. To characterize ligand solvation, we calculated the water coordination number around the chloride atom of EPI-002, denoted as $\{G\}$. For hydration at the binding site, we calculated the water coordination number around the sulfur atom of Cys404, denoted as $\{H\}$. In addition, high-density water regions within the binding site were identified as hydration spots $\{V\}$, with their geometric centers used to define supplementary descriptors through local water coordination analysis.

These descriptors $\{G\}$, $\{H\}$, $\{V\}$ were then incorporated into a multitask learning framework⁴⁶ to construct a prior CV for enhanced sampling. The multitask CV was trained using two datasets: (i) a labeled dataset derived from short unbiased MD simulations capturing bound and unbound states, and (ii) an unlabeled dataset obtained from OPES-explore¹⁶ trajectories that sample rare events and transition pathways. This formulation allows the CV to encode both structural differences between metastable states (via labeled data) and solvent-mediated transition mechanisms (via unlabeled data), thereby providing a practical CV for estimating ligand-protein binding free energies in disordered systems.

We next performed OPES-explore¹⁶ simulations to compute the binding free energy (ΔG) of EPI-002 to the nine metastable conformations of Tau-5 (Supplementary Fig. 7). For these simulations, we employed two CVs: (i) the distance z from the binding site, defined as the projection of the ligand's center of mass along the binding axis, and (ii) the multitask water CV described earlier. To ensure efficient sampling of binding/unbinding events, we applied a funnel-shaped

restraint potential^{73,74}, which confines the accessible volume of the unbound ligand and prevents it from drifting into bulk solvent. The entropic correction associated with the funnel restraint was computed analytically and incorporated into the final binding free energy. To minimize artifacts from protein conformational fluctuations, positional restraints were imposed to all C α atoms throughout the simulations.

Each trajectory exhibited multiple binding and unbinding events, ensuring adequate sampling of both bound and unbound states (Supplementary Figs. 9–17). Convergence of the binding free energy was rigorously verified across independent simulations (Supplementary Figs. 18–26). The resulting corrected ΔG , reported in Table 2, ranges from -3 to -5 kcal/mol across the nine metastable conformations, with conformation 6 showing the most favorable binding energy (-4.82 kcal/mol). This is consistent with previous computational estimates²² that reported $\Delta G = -5.26$ kcal/mol (corresponding to a K_D of 5.2 mM).

The observed weak binding affinity in the millimolar range contrasts with the low micromolar affinities reported experimentally^{75,76}. This discrepancy likely arises from the omission of covalent interaction in our calculation. Accurately quantifying the K_D of EPI-002 would require QM/MM MD simulations⁷⁷, which are beyond the scope of this study. Furthermore, prior NMR and ¹⁹F relaxation experiments also reported millimolar affinities for non-covalent IDP-ligand complexes⁷⁸, suggesting the biological relevance of weak binding in disordered systems.

Finally, we projected the two-dimensional FES onto the z coordinate and the multitask water CV (Supplementary Figs. 27–35). The resulting FES lacked well-defined basins, reflecting the limited accuracy of the multitask water CV. This outcome is expected, as a simple multitask framework does not guarantee the extraction of slow modes. Consequently, more rigorous MLCV algorithms are therefore required to capture the essential dynamical degrees of freedom governing IDP-ligand interactions.

Hydrogen bond network formation among EPI-002 and Tau-5 revealed by Deep-TICA

To elucidate the binding/unbinding mechanisms of EPI-002 with Tau-5, we retrained a set of Deep-TICA CVs using the trajectories from the OPES-explore simulations. To capture the evolution of ligand-protein interactions, we considered the distances between the aromatic moieties of the ligand and the aromatic side chains of Tau-5, for example, $\{dd_Tyr406_lig1\}$, defined as the distance between the aromatic center of Tyr406 and the ligand's distal ring (relative to the chlorine atom), and $\{dd_Tyr406_lig2\}$, defined as the distance to the proximal ring. Additional descriptors include the center-of-mass distance between the ligand and the binding site ($\{com\}$), as well as the distance between the sulfur atom of Cys404 and the chlorine atom of EPI-002 ($\{dd_Cys_lig\}$). These descriptors were designed to capture π - π stacking, noncovalent contacts, and potential reactivity at the chlorohydrin moiety. Using these inputs, we trained two Deep-TICA models: one to construct a Distance CV

Table 2 | The binding free energy ΔG (kcal/mol) of the nine metastable conformations of Tau-5 proteins and the EPI-002 compound. The mean and standard deviation are computed from the parallel OPES explore trajectories

Conformations	1	2	3	4	5	6	7	8	9
Mean	−3.39	−3.59	−3.33	−4.27	−3.23	−4.82	−3.36	−3.39	−4.16
Std	0.86	0.84	0.81	1.41	0.77	0.44	0.77	0.89	0.72

and the other trained using water coordination descriptors to yield the Water CV.

Projecting the two-dimensional FES onto the Deep-TICA Water and Distance CVs enabled us to resolve metastable binding states and assess descriptor contributions. Here, we focus on metastable conformations 2 and 6, which exhibited distinct binding behaviors and well-converged binding free energies (Supplementary Figs. 8 and 6). Complete analyses for all nine conformations are provided in Supplementary Figs. 36–49.

As shown in Supplementary Fig. 8, the FES of conformation 2 revealed three different binding poses (B, B1, B2) sharing similar overall geometries. Among them, pose B corresponds to the global free energy minimum, while B1 and B2 are higher in free energy by 0.369 kcal/mol and 2.359 kcal/mol, respectively. In pose B, EPI-002 forms hydrogen bonds with His382, Ala396, and Tyr445. In B1 and B2, subtle ligand rearrangements disrupt hydrogen bonding with Tyr445 and His382, respectively, destabilizing these states. Hydration analysis revealed that water regions $\{V_3\}$, $\{V_8\}$, $\{V_{16}\}$ (containing 4, 3, and 5 molecules, respectively) localize between EPI-002 and His382. During ligand dissociation, an intermediate state (I) emerges where protein–ligand hydrogen bonds are lost, and the pocket becomes fully hydrated, containing five water molecules. Three of these waters, residing in $\{V_3\}$ and $\{V_{16}\}$, form bridging hydrogen bonds between the chlorohydrin group of EPI-002 and His382. In the fully unbound state (U), the ligand is completely solvated, and the pocket returns to its apo form. Descriptor analysis identified $\{V_3\}$, $\{V_8\}$, $\{V_{16}\}$ as the strongest contributors to the Water CV, while the Distance CV was dominated by $\{dd\text{His382}\text{lig1}\}$, reflecting the central role of His382 in both direct and water-mediated interactions.

Figure 5 shows the FES of conformation 6, which exhibited the most favorable binding free energy among all nine metastable states. Two binding poses (B and B1) were identified, with B1 being 1.9 kcal/mol higher than B. In pose B, EPI-002 forms direct hydrogen bonds with Tyr406, His413, and Ala415. Two water molecules in $\{V_3\}$ form a bridging hydrogen bond network that links Cys404 with the ligand's secondary alcohol, extending to Tyr406 and Ala415. In contrast, pose B1 disrupts the Cys404–ligand hydrogen bond; instead, a water in $\{V_6\}$ forms a new bridge between Ser411 and the ligand's alcohol group, while $\{V_3\}$ waters maintain contacts with Tyr406.

During unbinding, two intermediates (I1 and I2) were observed. In I1, the ligand forms a hydrogen bond with Gly414, while waters in $\{V_3\}$ (four molecules) and $\{V_6\}$ (two molecules) create a complex bridging network with Cys404. In I2, the ligand interacts with Ser341, while three waters in $\{V_6\}$ form hydrogen bonds with Cys404, and the ligand transiently occupies the $\{V_6\}$ site.

Together, the Deep-TICA Water and Distance CV provide a mechanistic framework for understanding the EPI-002–Tau-5 binding and unbinding pathways. These results highlight several key insights: (i) His382 and Cys404 act as central anchoring residues through direct and water-mediated hydrogen bonds; (ii) bridging water molecules play an active role in stabilizing intermediate states and guiding the unbinding pathway; (iii) hydration-driven rearrangements of hydrogen-bond networks critically determine the stability of alternative binding poses. These insights advance our understanding of AR-NTD ligand recognition and provide quantitative guidance for designing next-generation antagonists targeting the AR IDP.

Discovery of AR-NTD antagonists based on identified druggable conformations

Building on our workflow, which identified druggable conformations of the AR-NTD and their interactions with the known modulator EPI-002, we applied this approach to discover antagonists through virtual screening (see Methods). The compound libraries from Specs (~200,000 compounds, downloaded in November 2023) and ChemDiv (ChemDiv 2021 release, ~200,000 compounds) were used. For each conformation, the top 1% of compounds from Specs and the top 0.1% from Chemdiv were retained based on their ranking. Compounds that were consistently enriched in at least three of the nine conformations were prioritized. Following drug-likeness filtering and clustering, 58 candidates were selected for biological evaluation, and their SMILES representations are provided in Supplementary Table 2.

Initially, these compounds were tested for antiproliferative activity in PCa cell lines with high AR expression, including LNCaP (expressing full-length AR) and 22Rv1 (expressing AR-V7). Six compounds exhibited notable inhibitory effects, each suppressing cell growth by over 35% in both cell lines (Fig. 6a). Among them, compound K53 demonstrated the most potent inhibitory effect (Fig. 6b). It not only outperformed the second-generation AR antagonist Enz in LNCaP cells but also effectively inhibited the proliferation of enzalutamide-resistant 22Rv1 and LNCaP–EnzR cells (Fig. 6c). Crucially, K53 displayed negligible cytotoxicity in the non-tumorigenic human prostate stromal cell line WPMY-1, comparable to Enz, and similarly exhibited minimal antiproliferative effects in non-tumorigenic fibroblast cells (Fig. 6d and Supplementary Fig. 51), indicating its high selectivity for cancer cells. Furthermore, to ensure the chemical integrity and purity of K53, full compound characterization data, including HPLC chromatograms as well as ^1H and ^{13}C NMR spectra, are provided in Supplementary Figs 55–57.

Mechanistic studies further confirmed the activity of K53. Surface plasmon resonance (SPR) assays using the AR-NTD–AF1 fragment revealed its direct binding with a dissociation constant (K_D) of 2.41×10^{-5} M (Fig. 6e). To assess the specificity of the observed interaction, reference compounds with well-characterized AR binding modes were incorporated into the SPR analysis. The AF1-targeting inhibitor EPI-002 exhibited a clear, concentration-dependent interaction with AF1, yielding a K_D of 6.28×10^{-5} M. In contrast, the LBD-directed antagonist enzalutamide failed to produce a dose-dependent SPR response, indicating that the assay selectively reports AF1 engagement (Supplementary Fig. 52A). A fluorescence polarization competition assay further indicated that K53 does not act through the canonical ligand-binding pocket. Whereas enzalutamide efficiently displaced a fluorescent ligand from the recombinant AR-LBD, K53 exhibited only weak and non-dose-dependent competition, even at high concentrations (Supplementary Fig. 52B). As an orthogonal validation in solution, ligand-observed ^1H NMR spectroscopy was performed. While K53 alone exhibited sharp, well-resolved resonances characteristic of a freely tumbling ligand, pronounced line broadening and signal attenuation were observed when K53 was combined with the AF1 fragment, indicating that K53 has affinity for AF1 (Fig. 6f). This interaction was further supported by WaterLOGSY experiments, which showed a characteristic signal sign inversion for K53 selectively in the presence of AF1, but not the AR DNA-binding domain (DBD) (Supplementary Fig. 53). Finally, target engagement was evaluated using a cellular thermal shift assay (CETSA) in LNCaP cells. Treatment with K53

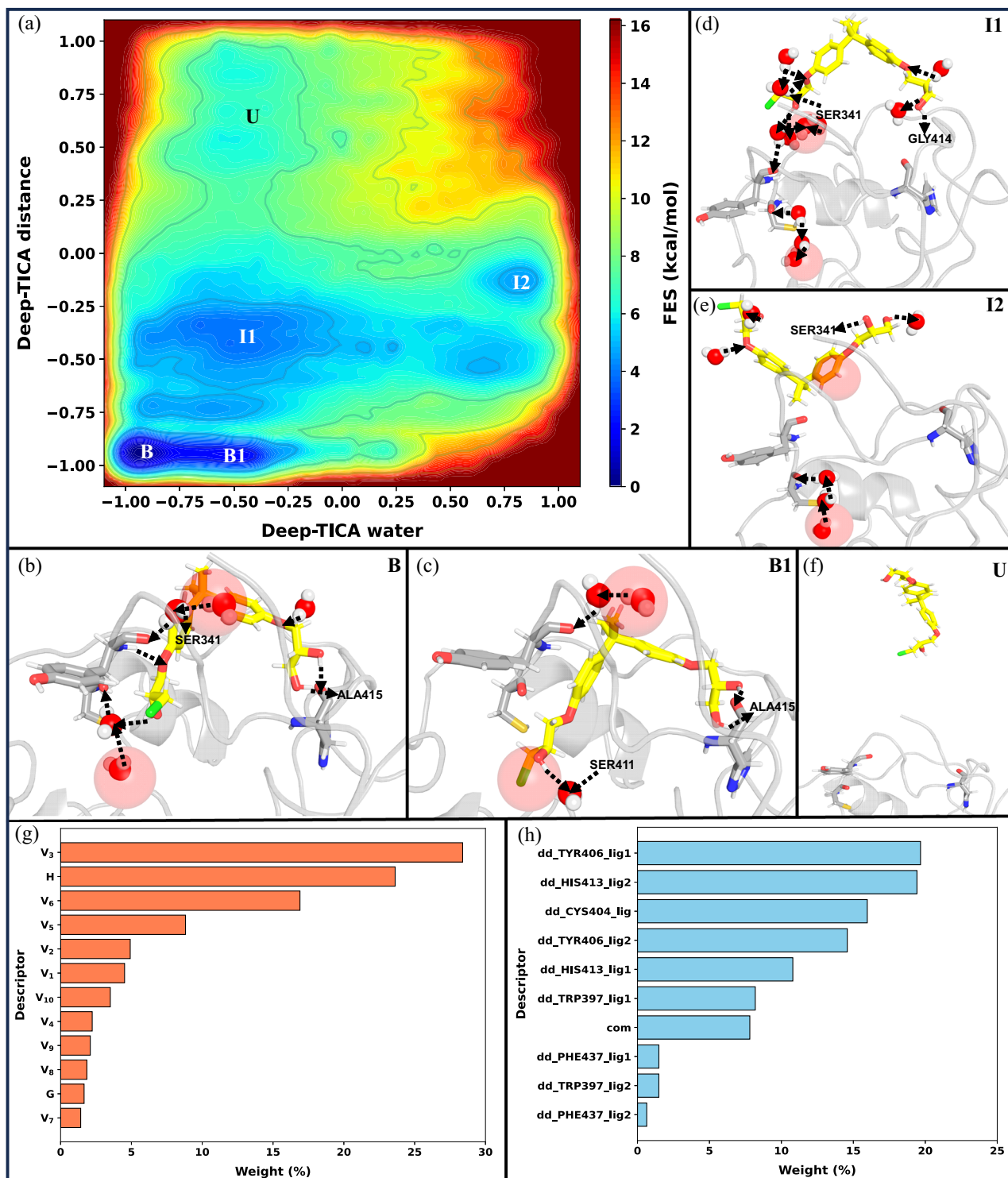


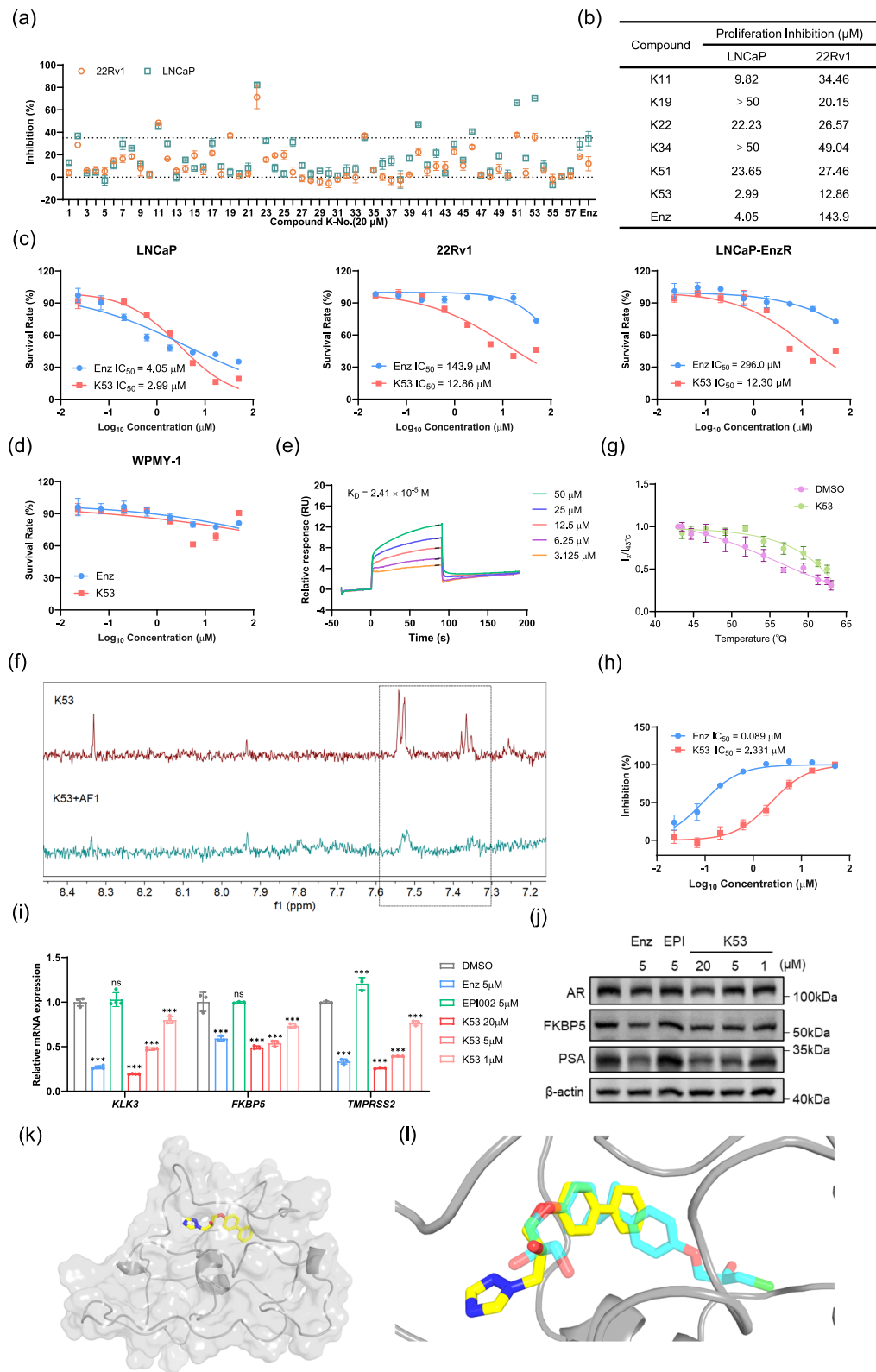
Fig. 5 | Deep-TICA analysis of the binding and unbinding processes between Tau-5 metastable conformation 6 and compound EPI-002. a 2D free energy landscape along the Deep-TICA water CV and Deep-TICA distance CV.

b–f Representative configurations of the relevant states are shown with a focus on the solvation patterns around the ligand, the aromatic residues within the binding site, and the binding poses. The highlighted residues, from top to bottom and left

to right, are Tyr406, Cys404, and His413. The transparent red spheres, arranged from top to bottom, represent the hydration sites V3 and V6. Except for the unbound (U) state, all configurations display water molecules surrounding the ligand and the binding pocket. **g, h** Descriptors' relative weights of the Deep-TICA Water CV and Deep-TICA Distance CV.

induced a reproducible rightward shift in the AR melting curve relative to the DMSO control, indicative of increased thermal stability and consistent with direct target engagement in cells (Fig. 6g). Collectively, these complementary biochemical, biophysical, and cellular assays support a specific interaction of K53 with the AR-NTD-AF1 region.

Given that AR functions as a nuclear hormone receptor whose oncogenic potential is mediated through transcriptional regulation, we further assessed the impact of K53 on AR-mediated transcriptional activity. Utilizing the LNCaP-ARR₂PB-eGFP cell line⁷⁹, K53 was found to dose-dependently inhibit dihydrotestosterone (DHT)-induced AR



activity, with an IC₅₀ of 2.331 μ M (Fig. 6h). Given the sequence similarity between the AR and other steroid receptors, we next examined the nuclear receptor selectivity of K53. Reporter assays showed that K53 did not antagonize the activity of progesterone receptor (PR), glucocorticoid receptor (GR), or mineralocorticoid receptor (MR) induced by exogenous hormones, respectively, nor did it activate PR or MR in the absence of hormones. These results indicate that K53 is largely specific to AR, without apparent cross-reactivity toward other

steroid receptors (Supplementary Fig. 54). Consistently, quantitative PCR (qPCR) analyses showed that K53 significantly downregulated the mRNA expression of canonical AR target genes including *KLK3* (PSA), *FKBP5*, and *TMPRSS2* in a concentration-dependent manner in LNCaP cells, whereas treatment with EPI-002 at 5 μ M exhibited no notable inhibitory effect (Fig. 6i). These findings were further corroborated at the protein level, indicating that K53 effectively disrupts AR signaling at both the transcriptional and translational levels (Fig. 6j).

Fig. 6 | Identification of AR-NTD antagonist K53. **a** Inhibition of cell proliferation in LNCaP and 22Rv1 cells by 58 virtual screening hits. Data are presented as mean $\pm$ SD from three independent replicates. **b** IC₅₀ values of six active compounds in LNCaP and 22Rv1 cells. **c** Dose–response curves of K53 and Enz in 22Rv1, LNCaP, and LNCaP-EnzR cells. Data are presented as mean $\pm$ SD from three independent replicates. **d** Cytotoxicity of K53 and Enz in non-tumorigenic WPMY-1 cells. Data are presented as mean $\pm$ SD from three independent replicates. **e** SPR binding profile of K53 with the AR-NTD-AF1 fragment. **f** The ligand-observed ¹H NMR analysis of K53 alone or in the presence of the AR AF1 fragment. **g** CETSA for assessing the binding of K53 to AR. Relative protein levels were quantified by Western blot (relative to β -actin) and normalized to the signal intensity at 43 °C. Data are presented as mean $\pm$ SEM from four independent biological replicates. Thermal denaturation curves were fitted using a Boltzmann sigmoidal model. Full blots are

provided in the Source Data files. **h** Inhibition of DHT-induced AR transcriptional activity by K53 and Enz in LNCaP-ARR₂PB-eGFP reporter cells. Data are presented as mean $\pm$ SD from three independent replicates. **i** qPCR analysis of AR target genes after 24 h treatment with indicated compounds. Data are presented as mean $\pm$ SD from three or four biological replicates. Significance was determined using one-way ANOVA with Dunnett's correction compared to DMSO. $p < 0.0001$ (***). **j** Western blot analysis of LNCaP cells treated with the indicated concentrations for 24 h. The data are representative of two independent experiments with similar results. Full blots and the replicate experiment are provided in the Source Data files. **k** The binding pose of compound K53 corresponds to metastable conformation 6 of Tau-5. **l** Three-dimensional conformational alignment of K53 (yellow) and EPI-002 (blue).

Structural modeling provided further insights. As shown in Fig. 6k, l, K53 binds within the hydrophobic pocket of metastable conformation 6 of Tau-5. The phenyl moiety anchors into the pocket through hydrophobic interactions, while the terminal triazole ring forms stable hydrogen bonds via its polar substituents. Importantly, the three-dimensional superposition reveals a pronounced structural resemblance between K53 and EPI-002: the conserved hydroxyl–ether–aromatic motif aligns closely in space, and the orientation of the hydrophobic aromatic rings shows a clear overlap. This strong structural similarity highlights a conserved structure–binding relationship, indicating that K53 remains within the EPI chemical scaffold while introducing modifications that may enhance activity and drug-like properties. Together, these interactions allow K53 to achieve a stable binding mode within Tau-5. This conserved structure–binding relationship suggests that K53 may retain functional features of EPI-like molecules while offering improved activity and drug-like properties.

Consistent with this structural analysis, OPES-based free energy calculations estimate a binding free energy of -8.33 kcal/mol for K53 in metastable conformation 6 (Supplementary Fig. S8), corresponding to a predicted K_D of approximately 8×10^{-7} M. This value is in reasonable agreement with the experimentally measured affinity of 2.41×10^{-5} M, supporting the validity of the identified binding mode. Together, these computational and experimental results indicate that K53 achieves a stable binding configuration within Tau-5. Furthermore, rational modification of K53, for example, incorporating a nucleophilic group to enable covalent interaction, represents a promising avenue to further enhance its potency.

Assessing the transferability of the workflow to c-Myc

To validate the general applicability of our workflow beyond AR-NTD, we applied it to the c-Myc system using the intrinsically disordered C-terminal fragment (N353-S437), demonstrating its transferability across distinct IDPs.

Using the same setup as for Tau-5, the system was modeled with a99SB-disp protein force field and the corresponding a99SB-disp water model. REST3 enhanced sampling was carried out over the 300–450 K temperature range for 1 μ s, using parameters identical to those employed for Tau-5. It is worth noting that, in this case, our objective was solely to sample metastable conformations, rather than to perform a comprehensive equilibrium characterization of the conformational ensemble as conducted for Tau-5. Analysis of the 300 K replica using an AE for dimensionality reduction revealed seven distinct metastable minima (Fig. 7a). Among these, metastable conformation 1 was selected for further investigation, as it provides a suitable structural basis for pocket opening and subsequent ligand docking (Fig. 7c). SWISH simulations were subsequently performed to promote pocket opening, into which the small-molecular inhibitor 10058-F4 was docked.

Following the same workflow, we first employed multi-task learning to construct a multi-task water CV. Subsequently, 150 ns OPES-explore

enhanced sampling was performed on the distance z and the multi-task water CV under a funnel-shaped restraining potential. The computed binding free energy of approximated -8.68 kcal mol⁻¹ (estimated $K_D \approx 0.4$ μ M) closely matches the experimentally observed micromolar-range affinity^{80,81} (Fig. 7d). In parallel, we trained a Deep-TICA water CV and projected the free energy landscape onto the z coordinate and the Deep-TICA water CV. The Deep-TICA water CV clearly distinguishes two distinct bound states, denoted as B and B1, while the z coordinate resolves the intermediate state I as well as the unbound state U (Fig. 7b).

Taken together, these results demonstrate that our workflow is not limited to AR-NTD but is readily transferable to other IDPs, exemplified here by the c-Myc system.

Discussion

In this work, we developed a broadly applicable enhanced sampling workflow that integrates MLCVs with generalized ensemble and importance sampling techniques to investigate druggable conformations of IDPs and their ligand-binding mechanisms. Using the AR-NTD and its modulator EPI-002 as a model system, we identified nine druggable conformations and revealed that ligand recognition is mediated not only by π – π stacking but also by key aromatic residues. These residues participate in extensive hydrogen-bond networks involving both trapped water molecules and the ligand, highlighting the cooperative role of structured waters in IDP–ligand interactions. Moreover, structure-based virtual screening against the identified conformations led to the discovery of the AR-NTD antagonist K53, whose activity was experimentally validated. This marks the first breakthrough in rational drug design targeting the intrinsically disordered AR-NTD.

Despite these advances, our workflow warrants further refinement. Future enhancements include incorporating machine learning committor function^{82,83} and replacing conventional feedforward architectures with graph neural networks⁸⁴. In addition, while EPI-002 is known to act via covalent binding, our study focused only on its noncovalent interactions. As a result, the chemical steps underlying covalent bond formation are not explicitly described here. Incorporating QM/MM calculations⁷⁷ or hybrid quantum-classical sampling schemes into the workflow would enable more accurate characterization of covalent reaction pathways and provide a more complete mechanism description of AR-NTD–ligand interactions.

More broadly, the workflow presented here is not limited to the androgen receptor but is readily extensible to other IDPs and challenging drug-discovery targets, such as c-Myc, tau, and related disordered regulatory proteins. Overall, our findings highlight the power of enhanced sampling as a lens to capture the dynamic interplay between ligands and disordered proteins. The growing body of research on complex biological systems continues to emphasize the importance of atomistic simulations and statistical approaches in advancing biophysics. We believe that combining machine learning with enhanced sampling provides a natural and effective paradigm for exploring the conformational dynamics of IDPs and guiding the rational design of next-generation therapeutics.

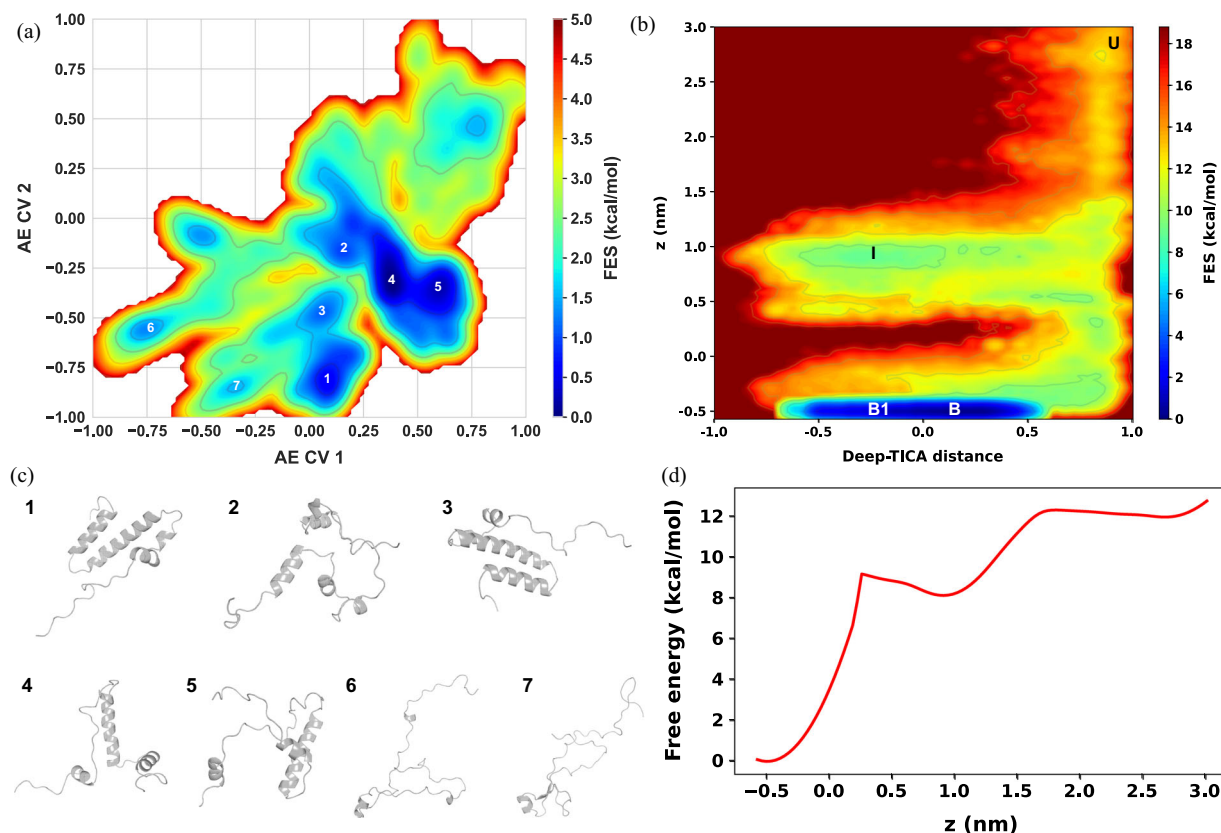


Fig. 7 | Application of the enhanced sampling workflow to the c-Myc system. **a** The FES of c-Myc was analyzed using the AE CV and the 3D structure of its nine metastable conformations. **b** Two-dimensional free energy landscape projected onto the Deep-TICA water CV and the z coordinate resolves for the binding of

10058-F4 to c-Myc. **c** Representative three-dimensional structures of the metastable conformations identified from the FES in panel (a). **d** One-dimensional free energy profile of 10058-F4 binding to c-Myc along the z coordinate.

Methods

All MD simulations were conducted using GROMACS 2022.6⁸⁵, integrated with the PLUMED 2.9.0⁸⁶ plugin and PyTorch library 2.2.1⁸⁷. The initial conformation of Tau-5 (residues 341–446) in the AR-NTD was predicted using AlphaFold2⁵⁵, with ACE and NME capping the termini. The a99SB-*disp* protein force field⁵⁶ was used for the Tau-5 protein, the a99SB-*disp* water model⁵⁶ was used for water molecules, and the GAFF force field⁸⁸ with RESP charges⁸⁹ was applied to characterize EPI-002. The steepest descent minimization algorithm was used for energy minimization. The time step was set to 2 fs. Bond lengths and angles for protein and ligand were constrained using the LINCS⁹⁰ algorithm, while water constraints were applied using the SETTLE⁹¹ algorithm. The temperature was maintained at 300 K under the NVT ensemble using the velocity rescale thermostat⁹² with a time constant of 2000 ps. The pressure was set to 1 bar under the NPT ensemble using the Parrinello-Rahman barostat⁹³, and the velocity rescale thermostat⁹² was used to control the temperature at 300 K with a time constant of 2000 ps. During the equilibration process, position restraints were applied to all protein-heavy atoms. The PME algorithm was used for electrostatics with a grid spacing of 1.2 nm. A cutoff length of 1.2 nm was used to calculate van der Waals interactions. The box for the REST3 simulation of Tau-5 is a cube with a side length of 12.5 nm, while the boxes for the other simulations are cubes with a side length of 10 nm. The boxes were solvated with water molecules, and sodium and chloride ions were added to neutralize the charge. All ML-CV training and analysis were performed using the mlcolvar⁴⁶ library.

Simulations of the Tau-5 protein

REST3 simulations. The REST3 algorithm⁴³ was implemented for microsecond-scale simulations, with replica exchanges occurring

every 2 ps. All protein atoms were selected as the solute region. The solute temperature was scaled using 8 replicas within the 300–450 K range. Each replica was simulated for 4.1 μ s, yielding a cumulative simulation time of 32.2 μ s. Further details of the parameter settings for the REST3 implementation are provided in Supplementary Table 1.

Blocking error analysis. Statistical errors for the REST3 simulations were analyzed using blocking analysis, with optimal block size determined as described in the previous study⁵⁸. The analysis was performed using the pyblock Python package. In this process, the MD trajectory is divided into a given number of equally sized “blocks”. For each block, time-averaged values of observables are computed. The standard error of these block-averaged values across all blocks is then calculated and adopted as the statistical error estimate.

Maximum-entropy ensemble reweighting. The NMR chemical shifts of the 300 K replica from the REST3 simulations of Tau-5 were calculated using SPARTA +⁶². A maximum entropy reweighting algorithm^{64,65} was applied with α NMR chemical shifts as constraints. The α NMR chemical shift predictions were performed using a Gaussian error model with a standard deviation of $\theta = 1.73$ ppm.

S α α -helical order parameter. The α -helical order parameter (S_α) measures the similarity of all six-residue segments to an ideal helical structure ($\phi = -57^\circ$, $\psi = -47^\circ$)⁵⁹:

$$S_\alpha = \sum_{i=1}^N \frac{1 - \left(\frac{\text{RMSD}_{\alpha_i}}{r_0} \right)^8}{1 - \left(\frac{\text{RMSD}_{\alpha_i}}{r_0} \right)^{12}}$$

where the sum is over N consecutive six-residue segments, RMSD_α^i represents the $\text{C}\alpha$ -RMSD between the ideal α -helix geometry and the six-residue segment (from residue i to residue $i+5$), with $r_0=1.0$ Å. When $r_0=1.0$ Å, six-residue segments with $\text{RMSD}_\alpha < 0.5$ Å contribute approximately 1.0 to the total S_α ; segments with $\text{RMSD}_\alpha = 1.1$ Å contribute approximately 0.5 to the total S_α ; and segments with $\text{RMSD}_\alpha > 3.0$ Å contribute approximately 0 to the total S_α .

Design of AE distance CV for Tau-5. To analyze the FES of the Tau-5 protein trajectory, AE CVs^{28,29} were employed to capture its complex dynamic transitions. From the 300 K REST3 trajectory, the $\text{C}\alpha$ - $\text{C}\alpha$ distance between R1 and R2, R2 and R3, and R1 and R3 within Tau-5 were selected as descriptors, generating 1899 $\text{C}\alpha$ - $\text{C}\alpha$ distance pairs that served as the input features for training the MLCVs. The neural network architecture comprised seven layers with 1899, 64, 64, 2, 64, 64, and 1899 nodes, with the shifted softplus as an activation function. The training was performed using the ADAM optimizer⁹⁴ with a learning rate of 1.0×10^{-4} and a mean squared error (MSE) loss function. Following the training process, the bottleneck layer of the AE was extracted and used as the CV.

SWISH Simulations. For each metastable conformation, we performed SWISH⁴⁵ simulations following a standardized protocol. Each system comprised four parallel replicas with a scaling factor (λ) uniformly distributed between 1.00 and 1.15, simulating 50 ns for each replica. Benzene at a concentration of 1 M was used as a cosolvent, with both the benzene molecules and the protein system thermally coupled to the same temperature group. By visually inspecting the trajectory at $\lambda=1.0$, we identified a conformation for each metastable state where the binding pocket was opened.

Molecular docking. The Protein Preparation Workflow module in Schrödinger was utilized to optimize any unreasonable regions in the protein structure. The ligand EPI-002 was prepared using the LigPrep module. Glide^{66,67} was employed to dock EPI-002 into the binding pockets of each metastable conformation. Then, the Covalent Docking module^{68,69} in Schrödinger's was used to perform covalent docking of EPI-002 into each metastable conformation. The reaction type was specified as a nucleophilic substitution, with Cys404 designated as the reactive residue.

Simulations of the Tau-5 protein complexed with the EPI-002 ligand

Design of multitask water CV. Our neural network architecture comprises a four-layer topology with 64, 16, 4, and 1 nodes. All hidden layers employ the shiftplus activation function. The descriptors consist of a set of water coordination numbers ($\{G\}$, $\{H\}$, $\{V\}$). We performed 30 ns unbiased simulations for both the bound and unbound states to generate the labeled data, and used OPES explore¹⁶ to sample the binding and unbinding times once as the unlabeled data. In the auxiliary TDA loss function²⁷ of the multitask CV⁴⁶, the target means and width of distributions corresponding to the two states (metastable A, B) were selected as $\mu_A = -10.00$, $\mu_B = 10.0$, and $\sigma_A = 1.0$, $\sigma_B = 1.0$. The weight factor of the auxiliary TDA loss was set as 1.0. The training was performed with the ADAM optimizer⁹⁴ with a learning rate of 1.0×10^{-5} .

The funnel restraint. In the OPES explore¹⁶ simulations, funnel restraints^{73,74} were applied to the systems. The funnel geometry confines the ligand to a cylindrical volume above the binding site in the dissociated state. As the ligand approaches the binding site, the funnel constraint gradually broadens to avoid obstructing binding events. The parameters defining the shape of the funnel constraint are provided in the PLUMED file.

Due to the funnel confinement, the free energy difference between the bound state B and the true unbound state U that we

extract from enhanced sampling simulations requires correction. It can be calculated via:

$$\Delta G = -\frac{1}{\beta} \log \left(C_0 \pi R_{\text{cyl}}^2 \int_B dz \exp(-\beta(W(z) - W_U)) \right) \quad (7)$$

where $\beta = 1/k_b T$, $C_0 = 1/1660$ Å⁻³ is the standard concentration, z is the coordinate along the axis of the funnel, $W(z)$ is the free energy along the funnel axis, and W_U is the reference free energy in state U .

OPES Explore simulations. For each EPI-002/Tau-5 metastable conformation, three independent OPES explore¹⁶ simulations were carried out: 300 ns for conformations 2 and 6, 160 ns for conformation 9, and 200 ns for the rest. During the simulations, bias potentials were applied to both the z and the multitask water CV. All simulations were conducted using a barrier estimate of 40 kJ mol⁻¹.

Design of Deep-TICA water CV and Deep-TICA distance CV. The Deep-TICA training strategy follows the protocol proposed by Bonati et al.³⁰, implementing a neural network architecture with four sequential layers utilizing the ReLU activation function. For the Water CV, the descriptors include a set of water coordination numbers ($\{G\}$, $\{H\}$, $\{V\}$), as described in the Results section. The Distance CV employs a set of ligand-protein interatomic distances, also detailed in the Results section. The training trajectories were derived from the OPES explore simulations. We employed the ADAM optimizer⁹⁴ with a learning rate of 1.0×10^{-4} . The loss function incorporated two TICA eigenvalues. The Deep-TICA model was trained with a lag time of 1, and the resulting CVs were used to analyze the FES of the OPES to explore biased trajectories.

Virtual screening and activity validation

Virtual screening. The nine metastable conformations of Tau-5 were used to generate a receptor network using the receptor network generation module in Schrödinger. Virtual screening against the Specs database was performed using the Glide^{66,67} docking program. Initially, ligands were prepared using the ligPrep module in Schrödinger, generating all possible protonation states at $\text{pH} = 7.0 \pm 2.0$ for two compound libraries: Specs and Chemdiv. For the Specs database, the ligand dataset was screened using the Glide SP scoring, and the top 20,000 compounds were redocked using the Glide XP scoring for higher accuracy. These compounds were then ranked based on their docking scores. The top 2000 compounds (top 1%) were retained. If a compound ranked in the top 1% for more than four conformations, it was kept, resulting in 215 compounds. These compounds underwent drug-likeness filtering using the ADMET⁹⁵⁻⁹⁷ module of Drugflow⁹⁸ and were clustered into 10 groups using the linkage clustering algorithm. A representative subset of compounds from each group was selected, resulting in a final set of 40 compounds. For the ChemDiv database, the top 200,000 compounds were initially selected using HTVS, followed by Glide SP screening to retain the top 20,000 compounds, and then redocked with Glide XP for higher precision. If a compound was ranked in the top 1% for more than three conformations, it was kept, resulting in a total of 161 compounds. These were filtered for drug-likeness using the ADMET⁹⁵⁻⁹⁷ module of Drugflow⁹⁸, and clustered into 10 groups using the linkage clustering algorithm. Representative compounds were selected from the clustering results, yielding a final set of 21 candidates. In total, 61 compounds were identified from both libraries, of which 58 were purchased, excluding those unavailable from suppliers.

Cell culture and reagents. LNCaP, 22Rv1, LNCaP-ARR2PB-eGFP, LNCaP-EnzR, NIH-3T3 and WPMY-1 cell lines were used in this study. The LNCaP-EnzR cell line was previously established and characterized by our group⁹⁹. The LNCaP-ARR2PB-eGFP reporter cell line, used to

assess AR transcriptional activity, was constructed and validated as previously described by stably introducing an androgen-responsive ARR₂PB promoter driving the expression of enhanced green fluorescent protein (eGFP)⁷⁹. WPMY-1 and NIH-3T3 were cultured in DMEM supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin. All prostate cancer cell lines were maintained in RPMI-1640 medium with 10% FBS and 1% penicillin-streptomycin at 37 °C in a humidified incubator with 5% CO₂. The Dihydrotestosterone (DHT) was purchased from Solabio, and Enz from MedChemExpress; both were dissolved in dimethyl sulfoxide (DMSO) to prepare stock solutions. A library of 58 structurally diverse candidate compounds for screening was purchased from TargetMol and prepared in DMSO at a final stock concentration of 20 mM for subsequent in vitro assays.

MTT Assay for cell proliferation. Cell viability was evaluated using the MTT colorimetric assay. LNCaP, 22Rv1, LNCaP-EnzR, NIH-3T3, and WPMY-1 cells were plated in 96-well plates at a density of 3000 to 5000 cells per well and incubated overnight to allow for cell attachment. The following day, cells were exposed to gradient concentrations of K53 or Enz for 72 hours, with LNCaP cells subjected to an extended 120 h treatment. At the end of the incubation period, 10 μ L of MTT reagent (5 mg/mL in PBS) was added to each well and incubated for 4 h at 37 °C to allow for formazan formation. After incubation, the medium was aspirated, and 100 μ L of a lysis solution containing 10% SDS and 0.1% HCl in PBS was added to solubilize the formazan crystals. The absorbance was measured at 570 nm using a Synergy H1 microplate reader (BioTek).

AR Transcriptional activity assay. The AR transcriptional activity was evaluated using the LNCaP-ARR₂PB-eGFP reporter cell line. Prior to compound treatment, cells were cultured in phenol red-free RPMI-1640 medium containing 5% charcoal-stripped serum (CSS) for five days to deplete endogenous androgens. Cells were then seeded into black-wall 96-well plates at a density of 3.0×10^4 cells per well and allowed to adhere for 24 h under standard conditions (37 °C, 5% CO₂). Subsequently, cells were treated with various concentrations of test compounds. 5 nM DHT was co-administered to activate AR signaling. After 72 h of treatment, eGFP fluorescence was measured using a Synergy H1 microplate reader (BioTek) with excitation and emission wavelengths set at 485 nm and 535 nm, respectively. Relative fluorescence intensity was used as a readout of AR transcriptional activity.

SPR Binding assay. Experiments were performed on a Biacore 8K system (GE Healthcare) using a CM5 sensor chip (#29104998; Cytiva). Recombinant AR-NTD-AF1 protein (Suzhou PLD Technology Co., Ltd.) was immobilized on the CM5 chip via standard amine coupling chemistry in PBST buffer (10 mM PBS containing 0.01% (v/v) P20 surfactant, pH 7.4). All measurements were performed at 25 °C. The analytes (K53, enzalutamide, and EPI-002) were injected at a flow rate of 30 μ L/min with contact and dissociation times of 120 s and 60 s, respectively. Each analyte was tested at a series of concentrations (3.125–50 μ M, two-fold serial dilution). A reference flow cell without immobilized protein was used, and data were processed using double referencing (reference surface subtraction and blank injection subtraction). Binding responses were recorded in multi-cycle kinetics mode. Data were analyzed using Biacore Insight Evaluation software (v5.0.18). Sensorgrams were fitted using a steady-state affinity model to derive K_D .

Cellular thermal shift assay (CETSA). LNCaP cells were seeded in 10 cm dishes and cultured in complete medium for 24 h, followed by treatment with the indicated compound at 30 μ M or DMSO as a control for 1 h. Cells were then detached, resuspended in PBS, and subjected to three freeze–thaw cycles using liquid nitrogen and a 25 °C water bath. Proteins were collected by high-speed centrifugation. The resulting

protein samples were heated at the indicated temperatures for 3 min in a PCR thermocycler, followed by cooling to room temperature for 3 min, and centrifuged again to isolate the soluble fraction for subsequent Western blot analysis. Band intensities were quantified by Western blotting (relative to β -actin) and normalized to the intensity of the 43 °C sample. Data are presented as the mean $\pm$ SEM of duplicate samples.

Nuclear magnetic resonance (NMR). ¹H NMR spectra of small molecules, either alone or in the presence of purified protein, were recorded on a Bruker 600 MHz spectrometer equipped with a CP2.1 QCI 600S3 probe. Standard ¹H experiments were acquired with 32 scans in a total volume of 500 μ L containing 500 μ M small molecule, either alone or in combination with 5 μ M protein, in 20 mM phosphate buffer prepared in 100% D₂O. WaterLOGSY experiments were performed using the ephogsygpn0.2 pulse sequence with 128 scans in a total volume of 500 μ L containing 500 μ M small molecule, either alone or with 5 μ M protein, in 20 mM phosphate buffer prepared in 90% H₂O/10% D₂O. All spectra were processed using TopSpin 4.5 with standard parameters.

Fluorescence polarization. Fluorescence polarization measurements were performed using the PolarScreen Androgen Receptor Competitor Assay Kit (#A15880; Invitrogen), following the manufacturer's protocol.

Nuclear receptor cross-reactivity analysis. To evaluate potential cross-reactivity with other steroid nuclear receptors, transcriptional reporter assays were performed for MR, PR, and GR. For MR and PR assays, HEK293T cells were plated into white 96-well plates at a density of 1×10^4 cells per well in DMEM supplemented with 5% CSS. Cells were transiently transfected with plasmids encoding MR or PR (70 ng), an MMTV-driven firefly luciferase reporter (24 ng), and a Renilla luciferase normalization construct (7 ng). After 12 h of transfection, cells were exposed to 100 nM aldosterone (for MR) or 100 nM progesterone (for PR) in the presence of the indicated concentrations of test compounds. Dual-luciferase signals were quantified using the Dual-Lumi™ Luciferase Assay Kit (#RG088M; Beyotime) in accordance with the manufacturer's instructions. For the GR transcriptional assay, HeLa-MMTV-Luc reporter cells were starved in white 96-well plates with DMEM with 5% CSS serum for hormone deprivation. Cells were then treated with 10 nM dexamethasone together with the indicated concentrations of compounds for 24 h. Firefly luciferase activity was measured using the Firefly Luciferase Assay Kit (#RG055M; Beyotime) following the manufacturer's protocol.

Western blot. LNCaP Cells were seeded in 12-well plates with 3×10^5 cells per well. After incubation at 37 °C with 5% CO₂ for 24 h, the cells were treated with the indicated compounds. After 24 h, the cells were lysed with lysis buffer (#P0013, Beyotime), and the total protein was separated by 10% SDS-PAGE (#PG113, Epizyme). Proteins were electrotransferred to poly(vinylidene fluoride) (PVDF) membrane (#IPFL00005, Millipore) and probed with primary antibodies specific to AR rabbit mAb (#5153, CST, 1:1000 dilution), PSA/CLK3 rabbit mAb (#5356, CST, 1:1000 dilution), β -actin Mouse mAb (#3700, CST, 1:1000 dilution) and FKBP51/FKBP5 rabbit pAb (#A3863, Abclonal, 1:3000 dilution). Antirabbit (#RGAR001, Proteintech, 1:5000 dilution) or antimouse (#RGAM001, Proteintech, 1:4000 dilution) secondary antibodies conjugated horseradish peroxidase (HRP) and enhanced chemiluminescence solution (#P0018S, Beyotime) were applied to visualize the proteins.

Quantitative real-time PCR (qPCR). LNCaP cells were seeded in 12-well plates at a density of 3×10^5 cells per well and incubated for 24 h, followed by treatment with K53, Enz, or EPI002 at the indicated

concentrations for 30 h. Total RNA was extracted using the EZ-10 DNA away RNA Mini-Preps Kit (#B618133, Sangon Biotech) according to the manufacturer's protocol. First-strand cDNA was synthesized using a reverse transcription kit (#11141ES10, YEASEN). qPCR was performed using SYBR Green Master Mix (#11202ES03, YEASEN) on a real-time PCR system (QuantStudio 3, Applied Biosystems). The relative expression levels of *KLK3* (PSA), *FKBP5*, and *TMPRSS2* were quantified using the $2^{-\Delta\Delta C_t}$ method, with GAPDH as the internal control. The primer sequences (synthesized by TSINGKE Biological Technology, Hangzhou, China) are listed as follows:

Gene	Forward primer (5'–3')	Reverse primer (5'–3')
GAPDH	GAAGGTGAAGGTCGGAGT	CATGGGTGAATCATATTGGAA
PSA	GGTGACCAAGTTCATGCTGTG	GTGTCCTTGATCCACTTCCG
TMPRSS2	CACGGACTGGATTATCGACAA	CGTCAAGGACGAAGACCATGT
FKBP5	ACCCCTTGGCTGACTCAAACCTCG	CCTTGCTGCCTTCTGAACCTG

Application of the workflow to the c-Myc system

The computational protocol employed for the c-Myc system closely followed that established for Tau-5. Accordingly, only methodological differences specific to the c-Myc setup are described below.

Initial structure construction. The initial structure of c-Myc was generated using the tleap module of the Amber¹⁰⁰ software package. The structure was first energy-minimized and then subjected to a short high-temperature unfolding simulation of 100 ps at 600 K in vacuo under the NVT ensemble. A representative structure was selected from the unfolding trajectory and used as the starting structure for subsequent simulations.

Design of AE distance CV for c-Myc. From the 300 K REST3 trajectory, all C α –C α distances within the c-Myc protein were selected as descriptors, generating 3570 C α –C α distance pairs that served as the input features for training the MLCVs. The neural network architecture comprised seven layers with 3570, 64, 64, 2, 64, 64, and 3570 nodes, respectively, employing the shifted softplus activation function.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data generated in this study have been deposited in a public repository and are available as Source Data files associated with this paper. Input files and results of MD simulations, as well as the code used for the analysis, have been deposited in Zenodo (<https://doi.org/10.5281/zenodo.19599898>)¹⁰¹. The androgen receptor Tau-5 NMR chemical shifts used for trajectory reweighting are from Biological Magnetic Resonance Bank entry BMRB Entry 51479. Source data are provided in this paper.

Code availability

MLCVs were trained using *mlcvar*. All neural networks were trained using the PyTorch library 2.2.1, and CVs were integrated into MD simulations through PLUMED 2.9.0. Secondary structure populations from chemical shifts were estimated using Delta2D, and C α chemical shifts were predicted using SPARTA+. All the inputs and instructions to reproduce the results presented in this manuscript can be found in the GitHub repository at https://github.com/Kai-Zhu-2001/AR-NTD_MD and is archived on Zenodo at <https://doi.org/10.5281/zenodo.19613110>¹⁰². The Python script used to determine areas of high water density and their centers can be found at this link.

References

- Dyson, H. J. & Wright, P. E. Intrinsically unstructured proteins and their functions. *Nat. Rev. Mol. Cell Biol.* **6**, 197–208 (2005).
- Wright, P. E. & Dyson, H. J. Intrinsically Disordered Proteins in Cellular Signaling and Regulation. *Nat. Rev. Mol. Cell Biol.* **16**, 18–29 (2015).
- Habchi, J., Tompa, P., Longhi, S. & Uversky, V. N. Introducing protein intrinsic disorder. *Chem. Rev.* **114**, 6561–6588 (2014).
- Toth, G. et al. Targeting the intrinsically disordered structural ensemble of α -synuclein by small molecules as a potential therapeutic strategy for Parkinson's disease. *PLoS ONE* **9**, e87133 (2014).
- Baggett, D. W. & Nath, A. The rational discovery of a tau aggregation inhibitor. *Biochemistry* **57**, 6099–6107 (2018).
- Yu, C. et al. Structure-based inhibitor design for the intrinsically disordered protein c-Myc. *Sci. Rep.* **6**, 22298 (2016).
- Ruan, H. et al. Computational strategy for intrinsically disordered protein ligand design leads to the discovery of p53 transactivation domain I binding compounds that activate the p53 pathway. *Chem. Sci.* **12**, 3004–3016 (2021).
- Hénin, J., Lelièvre, T., Shirts, M. R., Valsson, O. & Delemotte, L. Enhanced sampling methods for molecular dynamics simulations [Article v1.0]. *Living J. Comp. Mol. Sci.* **4**, <https://doi.org/10.33011/livecoms.4.1.1583> (2022).
- Valsson, O., Tiwary, P. & Parrinello, M. Enhancing important fluctuations: rare events and metadynamics from a conceptual viewpoint. *Annu. Rev. Phys. Chem.* **67**, 159–184 (2016).
- Torrie, G. M. & Valleau, J. P. Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling. *J. Comput. Phys.* **23**, 187–199 (1977).
- Laio, A. & Parrinello, M. Escaping free-energy minima. *Proc. Natl. Acad. Sci. USA* **99**, 12562–12566 (2002).
- Barducci, A., Bussi, G. & Parrinello, M. Well-tempered metadynamics: a smoothly converging and tunable free-energy method. *Phys. Rev. Lett.* **100**, 020603 (2008).
- Darve, E. & Pohorille, A. Calculating free energies using average force. *J. Chem. Phys.* **115**, 9169–9183 (2001).
- Invernizzi, M. & Parrinello, M. Rethinking metadynamics: from bias potentials to probability distributions. *J. Phys. Chem. Lett.* **11**, 2731–2736 (2020).
- Invernizzi, M., Piaggi, P. M. & Parrinello, M. Unified approach to enhanced sampling. *Phys. Rev. X* **10**, 041034 (2020).
- Invernizzi, M. & Parrinello, M. Exploration vs convergence speed in adaptive-bias enhanced sampling. *J. Chem. Theory Comput.* **18**, 3988–3996 (2022).
- Swendsen, R. H. & Wang, J.-S. Replica monte carlo simulation of spin-glasses. *Phys. Rev. Lett.* **57**, 2607–2609 (1986).
- Hansmann, U. H. E. Parallel tempering algorithm for conformational studies of biological molecules. *Chem. Phys. Lett.* **281**, 140–150 (1997).
- Liu, P., Kim, B., Friesner, R. A. & Berne, B. J. Replica exchange with solute tempering: A method for sampling biological systems in explicit water. *Proc. Natl. Acad. Sci. USA* **102**, 13749–13754 (2005).
- Wang, L., Friesner, R. A. & Berne, B. J. Replica exchange with solute scaling: a more efficient version of replica exchange with solute tempering (REST2). *J. Phys. Chem. B* **115**, 9431–9438 (2011).
- Miao, Y. et al. Improved reweighting of accelerated molecular dynamics simulations for free energy calculation. *J. Chem. Theory Comput.* **10**, 2677–2689 (2014).
- Zhu, J., Salvatella, X. & Robustelli, P. Small molecules targeting the disordered transactivation domain of the androgen receptor induce the formation of collapsed helical states. *Nat. Commun.* **13**, 6390 (2022).

23. Appadurai, R., Nagesh, J. & Srivastava, A. High resolution ensemble description of metamorphic and intrinsically disordered proteins using an efficient hybrid parallel tempering scheme. *Nat. Commun.* **12**, 958 (2021).
24. Zhao, J. et al. EGCG binds intrinsically disordered N-terminal domain of p53 and disrupts p53-MDM2 interaction. *Nat. Commun.* **12**, 986 (2021).
25. Zhu, K. et al. Enhanced sampling in the age of machine learning: algorithms and applications. *Chem. Rev.* **126**, 671–713 (2026).
26. Bonati, L., Rizzi, V. & Parrinello, M. Data-driven collective variables for enhanced sampling. *J. Phys. Chem. Lett.* **11**, 2998–3004 (2020).
27. Trizio, E. & Parrinello, M. From enhanced sampling to reaction profiles. *J. Phys. Chem. Lett.* **12**, 8621–8626 (2021).
28. Chen, W. & Ferguson, A. L. Molecular enhanced sampling with autoencoders: On-the-fly collective variable discovery and accelerated free energy landscape exploration. *J. Comput. Chem.* **39**, 2079–2102 (2018).
29. Belkacemi, Z., Gkeka, P., Lelièvre, T. & Stoltz, G. Chasing collective variables using autoencoders and biased trajectories. *J. Chem. Theory Comput.* **18**, 59–78 (2022).
30. Bonati, L., Piccini, G. & Parrinello, M. Deep learning the slow modes for rare events sampling. *Proc. Natl. Acad. Sci. USA*. **118**, e2113533118 (2021).
31. Das, S., Raucci, U., Neves, R. P. P., Ramos, M. J. & Parrinello, M. Correlating enzymatic reactivity for different substrates using transferable data-driven collective variables. *Proc. Natl. Acad. Sci. USA*. **121**, e2416621121 (2024).
32. Rizzi, V., Bonati, L., Ansari, N. & Parrinello, M. The role of water in host-guest interaction. *Nat. Commun.* **12**, 93 (2021).
33. Ansari, N., Rizzi, V. & Parrinello, M. Water regulates the residence time of Benzamidine in Trypsin. *Nat. Commun.* **13**, 5438 (2022).
34. Ruiz Munevar, M. J. et al. Cation chloride cotransporter NKCC1 operates through a rocking-bundle mechanism. *J. Am. Chem. Soc.* **146**, 552–566 (2024).
35. Claessens, F. et al. Diverse roles of androgen receptor (AR) domains in AR-mediated signaling. *Nucl. Recept Signal* **6**, e008 (2008).
36. Chandrasekar, T., Yang, J. C., Gao, A. C. & Evans, C. P. Mechanisms of resistance in castration-resistant prostate cancer. *Transl. Androl. Urol.* **4**, 365–380 (CRPC).
37. Luo, J. Development of AR-V7 as a putative treatment selection marker for metastatic castration-resistant prostate cancer. *Asian J. Androl.* **18**, 580 (2016).
38. Ji, Y., Zhang, R., Han, X. & Zhou, J. Targeting the N-terminal domain of the androgen receptor: The effective approach in therapy of CRPC. *Eur. J. Med. Chem.* **247**, 115077 (2023).
39. Moigne, R. et al. EPI-7386 is a novel N-terminal domain androgen receptor inhibitor for the treatment of prostate cancer. *Ann. Oncol.* **30**, v189–v190 (2019).
40. De Mol, E. et al. EPI-001, A compound active against castration-resistant prostate cancer, targets transactivation unit 5 of the androgen receptor. *ACS Chem. Biol.* **11**, 2499–2505 (2016).
41. Myung, J.-K. et al. An androgen receptor N-terminal domain antagonist for treating prostate cancer. *J. Clin. Invest.* **123**, 2948–2960 (2013).
42. Zhu, J. & Robustelli, P. Covalent adducts formed by the androgen receptor transactivation domain and small molecule drugs remain disordered. *J. Chem. Inf. Model.* **65**, 6221–6237 (2025).
43. Zhang, Y., Liu, X. & Chen, J. Re-balancing replica exchange with solute tempering for sampling dynamic protein conformations. *J. Chem. Theory Comput.* **19**, 1602–1614 (2023).
44. Hinton, G. E. & Salakhutdinov, R. R. Reducing the dimensionality of data with neural networks. *Science* **313**, 504–507 (2006).
45. Oleinikovas, V., Saladino, G., Cossins, B. P. & Gervasio, F. L. Understanding cryptic pocket formation in protein targets by enhanced sampling simulations. *J. Am. Chem. Soc.* **138**, 14257–14263 (2016).
46. Bonati, L., Trizio, E., Rizzi, A. & Parrinello, M. A unified framework for machine learning collective variables for enhanced sampling simulations: mlcolvar. *J. Chem. Phys.* **159**, 014801 (2023).
47. Molgedey, L. & Schuster, H. G. Separation of a mixture of independent signals using time delayed correlations. *Phys. Rev. Lett.* **72**, 3634–3637 (1994).
48. Naritomi, Y. & Fuchigami, S. Slow dynamics in protein fluctuations revealed by time-structure based independent component analysis: The case of domain motions. *J. Chem. Phys.* **134**, 065101 (2011).
49. Keller, B. G., Prinz, J.-H. & Noé, F. Markov models and dynamical fingerprints: Unraveling the complexity of molecular kinetics. *Chem. Phys.* **396**, 92–107 (2012).
50. Schwantes, C. R. & Pande, V. S. Improvements in markov state model construction reveal many non-native interactions in the folding of NTL9. *J. Chem. Theory Comput.* **9**, 2000–2009 (2013).
51. Wu, H. et al. Variational Koopman models: Slow collective variables and molecular kinetics from short off-equilibrium simulations. *J. Chem. Phys.* **146**, 154104 (2017).
52. Lavery, D. N. & McEwan, I. J. Functional characterization of the native NH2-terminal transactivation domain of the human androgen receptor: binding kinetics for interactions with TFIIF and SRC-1a. *Biochemistry* **47**, 3352–3359 (2008).
53. Kumar, R., Betney, R., Li, J., Thompson, E. B. & McEwan, I. J. Induced α -helix structure in AF1 of the androgen receptor upon binding transcription factor TFIIF. *Biochemistry* **43**, 3008–3013 (2004).
54. Bielskutė, S. et al. Low amounts of heavy water increase the phase separation propensity of a fragment of the androgen receptor activation domain. *Protein Sci.* **30**, 1427–1437 (2021).
55. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
56. Robustelli, P., Piana, S. & Shaw, D. E. Developing a molecular dynamics force field for both folded and disordered protein states. *Proc. Natl. Acad. Sci. USA*. **115**, E4758–E4766 (2018).
57. Flyvbjerg, H. & Petersen, H. G. Error estimates on averages of correlated data. *J. Chem. Phys.* **91**, 461–466 (1989).
58. Wolff, U. & Collaboration, A. Monte Carlo errors with less errors. *Comput. Phys. Commun.* **156**, 143–153 (2004).
59. Pietrucci, F. & Laio, A. A collective variable for the efficient exploration of protein beta-sheet structures: application to SH3 and GB1. *J. Chem. Theory Comput.* **5**, 2197–2201 (2009).
60. Camilloni, C., De Simone, A., Vranken, W. F. & Vendruscolo, M. Determination of secondary structure populations in disordered states of proteins using nuclear magnetic resonance chemical shifts. *Biochemistry* **51**, 2224–2231 (2012).
61. Basu, S. et al. Rational optimization of a transcription factor activation domain inhibitor. *Nat. Struct. Mol. Biol.* **30**, 1958–1969 (2023).
62. Shen, Y. & Bax, A. SPARTA+: a modest improvement in empirical NMR chemical shift prediction by means of an artificial neural network. *J. Biomol. NMR* **48**, 13–22 (2010).
63. Huang, J. et al. CHARMM36m: an improved force field for folded and intrinsically disordered proteins. *Nat. methods* **14**, 71–73 (2017).
64. Cesari, A., Gil-Ley, A. & Bussi, G. Combining simulations and solution experiments as a paradigm for RNA force field refinement. *J. Chem. Theory Comput.* **12**, 6192–6200 (2016).
65. Cesari, A., Reißer, S. & Bussi, G. Using the maximum entropy principle to combine simulations and solution experiments. *Computation* **6**, 15 (2018).

66. Friesner, R. A. et al. Glide: a new approach for rapid, accurate docking and scoring. 1. Method and assessment of docking accuracy. *J. Med. Chem.* **47**, 1739–1749 (2004).
67. Halgren, T. A. et al. Glide: a new approach for rapid, accurate docking and scoring. 2. Enrichment factors in database screening. *J. Med. Chem.* **47**, 1750–1759 (2004).
68. Zhu, K. et al. Docking covalent inhibitors: a parameter free approach to pose prediction and scoring. *J. Chem. Inf. Model.* **54**, 1932–1940 (2014).
69. Toledo Warshaviak, D., Golan, G., Borrelli, K. W., Zhu, K. & Kalid, O. Structure-based virtual screening approach for discovery of covalently bound ligands. *J. Chem. Inf. Model.* **54**, 1941–1950 (2014).
70. Hummer, G. Under water's influence. *Nat. Chem.* **2**, 906–907 (2010).
71. Bellissent-Funel, M.-C. et al. Water determines the structure and dynamics of proteins. *Chem. Rev.* **116**, 7673–7697 (2016).
72. Schirò, G. et al. Translational diffusion of hydration water correlates with functional motions in folded and intrinsically disordered proteins. *Nat. Commun.* **6**, 6490 (2015).
73. Limongelli, V., Bonomi, M. & Parrinello, M. Funnel metadynamics as accurate binding free-energy method. *Proc. Natl. Acad. Sci. USA.* **110**, 6358–6363 (2013).
74. Raniolo, S. & Limongelli, V. Ligand binding free-energy calculations with funnel metadynamics. *Nat. Protoc.* **15**, 2837–2866 (2020).
75. Andersen, R. J. et al. Regression of castrate-recurrent prostate cancer by a small-molecule inhibitor of the amino-terminus domain of the androgen receptor. *Cancer Cell* **17**, 535–546 (2010).
76. Peng, S. et al. Regression of castration-resistant prostate cancer by a novel compound QW07 targeting androgen receptor N-terminal domain. *Cell Biol. Toxicol.* **36**, 399–416 (2020).
77. Senn, H. M. & Thiel, W. QM/MM methods for biomolecular systems. *Angew. Chem. Int. Ed.* **48**, 1198–1229 (2009).
78. Heller, G. T., Shukla, V. K., Figueiredo, A. M. & Hansen, D. F. Picosecond dynamics of a small molecule in its bound state with an intrinsically disordered protein. *J. Am. Chem. Soc.* **146**, 2319–2324 (2024).
79. Zhou, W. et al. Discovery of novel androgen receptor ligands by structure-based virtual screening and bioassays. *Genom. Proteom. Bioinform.* **16**, 416–427 (2018).
80. Michel, J. & Cuchillo, R. The impact of small molecule binding on the energy landscape of the intrinsically disordered protein C-Myc. *PLOS ONE* **7**, e41070 (2012).
81. Follis, A. V., Hammoudeh, D. I., Wang, H., Prochownik, E. V. & Metallo, S. J. Structural rationale for the coupled binding and unfolding of the c-Myc oncoprotein by small molecules. *Chem. Biol.* **15**, 1149–1155 (2008).
82. Kang, P., Trizio, E. & Parrinello, M. Computing the committor with the committor to study the transition state ensemble. *Nat. Comput. Sci.* <https://doi.org/10.1038/s43588-024-00645-0> (2024).
83. Trizio, E., Kang, P. & Parrinello, M. Everything everywhere all at once: a probability-based enhanced sampling approach to rare events. *Nat. Comput. Sci.* <https://doi.org/10.1038/s43588-025-00799-5> (2025).
84. Zhang, J. et al. Descriptor-free collective variables from geometric graph neural networks. *J. Chem. Theory Comput.* **20**, 10787–10797 (2024).
85. Abraham, M. J. et al. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **1**, 19–25 (2015).
86. Tribello, G. A., Bonomi, M., Branduardi, D., Camilloni, C. & Bussi, G. PLUMED 2: New feathers for an old bird. *Comput. Phys. Commun.* **185**, 604–613 (2014).
87. Paszke, A. et al. Automatic differentiation in pytorch. In *Proceedings of the 31st Conference on Neural Information Processing Systems*(2017).
88. Wang, J., Wolf, R. M., Caldwell, J. W., Kollman, P. A. & Case, D. A. Development and testing of a general amber force field. *J. Comput. Chem.* **25**, 1157–1174 (2004).
89. Bayly, C. I., Cieplak, P., Cornell, W. & Kollman, P. A. A well-behaved electrostatic potential based method using charge restraints for deriving atomic charges: the RESP model. *J. Phys. Chem.* **97**, 10269–10280 (1993).
90. Hess, B., Bekker, H., Berendsen, H. J. & Fraaije, J. G. LINCS: a linear constraint solver for molecular simulations. *J. Comput. Chem.* **18**, 1463–1472 (1997).
91. Miyamoto, S. & Kollman, P. A. Settle: An analytical version of the SHAKE and RATTLE algorithm for rigid water models. *J. Comput. Chem.* **13**, 952–962 (1992).
92. Bussi, G., Donadio, D. & Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **126**, 014101 (2007).
93. Parrinello, M. & Rahman, A. Polymorphic transitions in single crystals: A new molecular dynamics method. *J. Appl. Phys.* **52**, 7182–7190 (1981).
94. Kingma, D. P. & Ba, J. Adam: A method for stochastic optimization. *International Conference on Learning Representations (ICLR)* (2015).
95. Wu, Z. et al. Chemistry-intuitive explanation of graph neural networks for molecular property prediction with substructure masking. *Nat. Commun.* **14**, 2585 (2023).
96. Xiong, G. et al. ADMETlab 2.0: an integrated online platform for accurate and comprehensive predictions of ADMET properties. *Nucleic Acids Res.* **49**, W5–W14 (2021).
97. Dong, J. et al. ADMETlab: a platform for systematic ADMET evaluation based on a comprehensively collected ADMET database. *J. Cheminform.* **10**, 1–11 (2018).
98. Shen, C. et al. DrugFlow: An AI-driven one-stop platform for innovative drug discovery. *J. Chem. Inf. Model.* **64**, 5381–5391 (2024).
99. Chai, X. et al. Unraveling the efficacy of AR antagonists bearing *N*-(4-(Benzyloxy)phenyl)piperidine-1-sulfonamide scaffold in prostate cancer therapy by targeting LBP mutations. *J. Med. Chem.* **68**, 11962–11978 (2025).
100. Case, D. A. et al. The Amber biomolecular simulation programs. *J. Comput. Chem.* **26**, 1668–1688 (2005).
101. Zhu, K. et al. Targeting the intrinsically disordered AR-NTD through a machine learning-based enhanced sampling workflow. *Zenodo*<https://doi.org/10.5281/zenodo.19599898> (2026).
102. Zhu, K. Kai-Zhu-2001/AR-NTD_MD: Code for: Targeting the intrinsically disordered AR-NTD using a machine learning-based enhanced sampling workflow (v1.0). *Zenodo*<https://doi.org/10.5281/zenodo.19613110> (2026).

Acknowledgements

We thank J. Pan and D. Wu of the Research and Service Center, College of Pharmaceutical Sciences, Zhejiang University, for technical assistance with qPCR, and J. Zhu and Y. Wang for useful discussion. We thank Y. Liu of the Chemistry Instrumentation Center, Zhejiang University, for technical support with NMR. We thank S. Wu of the Analysis Center of Agrobiological and Environmental Sciences, Zhejiang University, for technical assistance with SPR analysis.

Author contributions

K.Z., H.W., J.Z., T.H., and D.L. conceived and designed the project. K.Z. developed the computational workflow and performed the subsequent MD simulations. H.W. conducted all wet-lab experiments. J.Z. provided overall scientific guidance for the project. R.H., L.J., H.Z., and Y.K.

contributed to data visualization and figure preparation. K.Z. and H.W. jointly wrote the initial draft of the manuscript, and J. Z., T.H., and D.L. participated in revising and approving the final version.

Funding

The authors disclose support for the research of this work from the National Key Research and Development Program of China [grant number 2024YFA1307501], the Jinhua Science and Technology Plan Projects [grant number 2023-3-077], the Central Guidance for Local Science and Technology Development Funds Project [grant number 2025ZY01022], and the National Natural Science Foundation of China [grant numbers 92370130, 82473843 and 82204279].

Competing interests

The authors declare the following competing financial interest(s): A Chinese patent application related to the AR-NTD antagonists described in this manuscript has been filed. The patent applicant is Zhejiang University. The listed inventors are Dan Li, Tingjun Hou, Kai Zhu, Huating Wang, Rong Sheng and Jintu Zhang. The application number is Chinese Patent Application No. 202511953361.2, filed on 23 December 2025. The application has been accepted by the China National Intellectual Property Administration and is currently pending. The application covers compounds with androgen receptor antagonist activity and their applications. There are no more competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-73863-x>.

Correspondence and requests for materials should be addressed to Tingjun Hou or Dan Li.

Peer review information *Nature Communications* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026